Results in Engineering 22 (2024) 102225

Contents lists available at ScienceDirect

Results in Engineering

journal homepage: www.sciencedirect.com/journal/results-in-engineering

Review article

A review on recent photovoltaic module cooling techniques: Types and assessment methods

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ARTICLE INFO

Keywords:
PV cooling
Classification of
Passive cooling
Active cooling
Combined cooling
Optimal operating temperature
Electrical efficiency
F_{TDED}
F_{ED}
R
F_{LSE}
F_{CE}
F_{MCE}
F_{CAE}

ABSTRACT

Solar energy has emerged as a standout alternative among the various types of renewable energies due to availability and minimal upfront expense in energy conversion. One of the most comprehensive methods to utilize solar energy potential is photovoltaic (PV) systems that apply a phenomenon called "photovoltaic effect". However, one major obstacle to the widespread use of PV is their optimal working temperature. The PV performance is based on heat distribution on surface of the panel that need to be decreased. From previous literature reviews on PV cooling, it is found there's insufficient attention given to discussing the vital and standardized parameters necessary for evaluating the efficiency of cooling techniques for PV systems. Hence, in this article, several updated studies on PV cooling technology are discussed including passive, active and combined cooling. Besides that, the performance assessment methods are also elaborated including temperature-dependent PV efficiency difference factor (F_{TDED}), temperature dependent photovoltaic power difference factor (F_{TDPD}), PV power difference factor (F_{ED}), power ratio (R), PV cooler lifespan efficacy factor (F_{LSE}), production cost effectiveness factor (F_{CE}), modified production cost effectiveness factor (F_{MCE}), and cost and area effectiveness factor (F_{CAE}). It was found that the PV cooling has some major challenges to be addressed in future studies including cost, maintenance requirement, energy consumption especially for active cooling methods and performance under extreme conditions. Hence, recommendations are given for the purpose of tackling some of these challenges.

Nomenclature

A	Area, (m^2)	P	power, (W)
β	efficiency decrease per unit increase in temperature, ($°C^{-1}$)	R	ratio, dimensionless
F	factor, dimensionless	T	temperature, (°C)
I	solar irradiance, (W/m^2)	Y	cost of 1 W of PV, ($)
L	lifespan, (Year)	Z	cost of the cooling technique, ($)
n	number of solar cells	P	power, (W)
STC	standard test condition		
Subscripts			
CE	cost effectiveness production cost effective	max	maximum output power
		MCE	modified cost effectiveness
		min	minimum
cell	photovoltaic cell	PV	photovoltaic module

(*continued on next column*)

(*continued*)

converted	converted area (m2)	PVCT	photovoltaic module with a cooler
ED	PV power difference	ref	reference
fc	forced convection	TDED	PV efficiency difference factor
		TDPD	PV power difference factor
LSE	lifespan effectiveness		

1. Introduction

An unprecedented surge in the demand for energy across numerous industries has been foretold by the steady expansion in both the population and economic activities. The fossil fuels had developed due to the steadily expanding energy demand for home and industrial uses. Thus, the widespread use of non-renewable fossil fuels has increased the

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https://doi.org/10.1016/j.rineng.2024.102225
Received 23 February 2024; Received in revised form 18 April 2024; Accepted 3 May 2024
Available online 10 May 2024
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atmospheric carbon dioxide emissions while also accelerating the depletion of these resources [1]. Because fossil fuels release greenhouse gases into the atmosphere, which contribute to global warming and climate change, the conventional use of fossil fuels has raised significant environmental concerns. Various technologies have been utilized to address energy-related issues and reduce reliance on fossil fuels [2]. On the other hand, the urgency of energy generation shall be transited from fossil fuel to renewable energy in order to reach and accomplish the Sustainable Development Goals (SDGs) of the UN and the Paris Agreement [3,4]. Therefore, it is imperative to look into and develop renewable energy alternatives that can take the place of traditional fossil fuels [152].

Renewable energy included solar, wind, hydro, geothermal, wave, biomass, hydrogen [153]. Renewable energy typically requires operational area per unit of energy than mining commodities or nuclear energy, though this varies depending on technology and conditions [5,6]. The utilization of land for agriculture is limited by the fact that most renewable energy sources require sizable footprints in order to produce power on a greater scale [7]. Among the various types of renewable energies, solar energy has emerged as a standout alternative due to numerous benefits such as this immense clean energy is available every day, so it can be harnessed and the majority of solar energy conversion methods have a minimal upfront expense [8].

One of the most comprehensive methods to utilize solar energy potential is photovoltaic (PV) systems [154]. It is also known as photo-electric conversion semiconductors which are used in PV modules to directly convert absorbed solar radiation into electrical power. Further improvement in PV conversion efficiency and cost reduction are needed to support PV development. The most widely used Silicon (Si) cells have conversion efficiencies that almost match their theoretical maximum. To fully achieve the potential of solar energy and ordinary photovoltaics, there are challenges that need to be overcome such as land use, intermittency, storage facilities, and integration into current energy networks. Given that solar energy is now viewed as a sustainable alternative to traditional energy sources, thorough research projects that include regular reviews and follow-ups of existing studies are essential [9]. The development in capacity of PV output has been steadily growing over the last 10 years. In 2018, it produced 600 TWh of power, or 2.4 % of the world's total electricity. By 2025, it is predicted to have reached 22 % of the total, with the potential to reach 70 %. According to the International Energy Agency Photovoltaic Power Systems Programme (IEA PVPS) report, the total PV capacity worldwide was 945,7 GW in 2022 [155]. Nonetheless, despite recent advancements in PV cell efficiency, the output of electricity intensity is still less than heat-producing facilities [10,11]. However, one major obstacle to the widespread use of PV is their optimal working temperature. The PV performance is based on the surface's heat that need to be stable against the increasing by radiation of the sun at PV outer layer. PV cooling is one of the easiest methods to improve its performance, but it's a challenging issue in the experimental work and field application [12].

The high temperature effects may cause the negative issues on I–V characteristic of PV. PV modules showed the best performance at cooler temperature and degrade as temperature warm up [13]. PV modules' current output increases in high temperatures [14]. On the other hand, the voltage drops at the PV module's output terminals as the temperature rises. The total performance and power production of the PV system may be impacted by the drop voltage [156]. Finally, because of the high temperatures, the overall power output decreased as a result of the reduced voltage output. This power loss affects the efficiency and performance of the PV system [15].

To encourage further advancements in the PV cooling technology, the study in the massive review articles detailing current PV cooling techniques shall be essential. Researchers have been actively looking into different cooling techniques to boost the effectiveness and cost-efficiency of PV module. Due to this interest, a large number of studies exploring PV cooling methods have been conducted. Several comprehensive review studies have detailed discussion on particular methods, in example on Phase Change Materials (PCM) [157], on nanofluid [16], on water cooling [17], on water and PCM cooling [158], on air cooling [18], on thermoelectric cooling [19], on passive cooling [20], on uniform cooling [159], on cooling techniques without thermoelectric and evaporative cooling, on active, passive, and combined cooling [21], on fin cooling [22], on trend of PV cooling, on advanced review [23], and on various PV cooling [24].

1.1. Motivation of the current work

The common review on PV cooling field usually put attention in the result and performance analysis. But the important and standard parameters to assess how well PV cooling works in term of economic effectiveness, power performance, and lifespan are rarely discussed. Numerous variables can impact the PV cooling system's performance including temperature-dependent PV efficiency difference factor (F_{TDED}), temperature dependent photovoltaic power difference factor (F_{TDPD}), PV power difference factor (F_{ED}), power ratio (R), PV cooler lifespan efficacy factor (F_{LSE}), production cost effectiveness factor (F_{CE}), and modified production cost effectiveness factor (F_{MCE}), and cost and area effectiveness factor (F_{CAE}). All of these parameters have studied and published previously.

In this review article, several updated researches and works in PV cooling technology will be discussed. Some examples and explanation of each PV cooling classification and results from previous researches have been done to give the brief and clear information to the readers. There are also brief performance assessment methods to evaluate the performance of the PV cooling method in different aspects, such as economic, power, efficiency, and lifespan. These assessment methods are F_{TDED}, F_{TDPD}, F_{ED}, R, F_{LSE}, F_{CE}, F_{MCE}, and F_{CAE}. These two main topics will help the researchers to identify and analysis the development of PV cooling technology. The major challenges in PV cooling are identified and recommendations for future work to tackle these challenges are given.

2. PV cooling classification

The high operating temperature of photovoltaics is one of the key problems with using them to generate electricity. In this respect, thermally regulated photovoltaic systems have attracted a lot of attention lately due to their higher productivity and efficiency when compared to regular PV modules. Instead of turning solar radiation into energy and squandering it as heat, these methods gather it into a heat-storage medium [25]. In order to keep the temperature of the solar panels within the working range specified by the manufacturer, heat removal from the panels using uniform cooling techniques is necessary. Numerous cooling methods are used to improve PV panel performance and regulate thermals [26].

Fig. 1 displays the I–V curve characteristics of PV panels at various temperatures and at 1000 W/m^2 solar irradiation. This data was collected using the Pvsyst program and takes into account a 300 W module. Temperature (T) has a significant impact on voltage, and as temperature rises, voltage (V) decreases. As temperature drops, PV current (I) marginally reduces while PV voltage sharply increases. Maximum power output is observed for temperature rises of up to 25 °C. Continuing with the gradual rise in temperature [27].

Changes in surface temperature can be caused by a variety of external climate conditions, such as sunlight, wind direction, moisture content, atmospheric temperature, and concentrated dust. Because adjusting the other parameters creates more difficulties, lowering the operating temperature can boost efficiency. For example, photovoltaic panels on building facades, which are non-directional, vertical surfaces, are designed with solar radiation as an unpredictable component. In an attempt to reduce the issue of temperature rise and boost solar efficiency, numerous cooling techniques have been tried and recorded in a variety of publications [161]. The performance of PV modules in terms

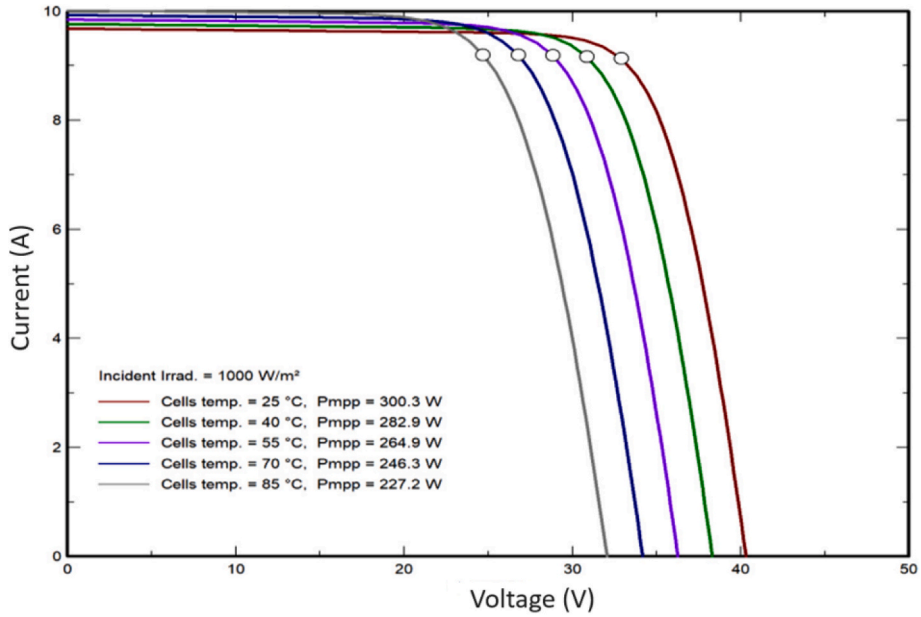


Fig. 1. I–V properties of a photovoltaic panel at varying temperatures at 25 °C of ambient temperature [160].

of electrical production may be impacted by variations in temperature and the intensity of solar radiation over an extended period of time. Only 20 % of solar radiation is converted into useful electricity; the other amount is lost as heat, which raises the module's working temperature. Thus, suitable cooling methods can be applied to enhance PV performance [28].

PV panels can be cooled using a variety of techniques based on either passive or active processes. The following sections offer a comprehensive and up-to-date overview of cooling techniques. These results are classified into three heat transfer modes: convective, conductive, and radiative cooling [162]. Many cooling techniques have been studied in an attempt to lower the temperature of the PV module by removing the heat it generates. The two main methods for controlling the temperature of PV modules are passive and active. Active cooling requires additional components (a blower or pump) to move the coolant fluid [163]. Future research on PV cooling must address several significant challenges, including cost, maintenance needs, energy consumption, especially for active cooling methods, and performance in extreme conditions [13].

Researchers have studied cooling techniques both theoretically and empirically, dividing them into three main groups and eight subcategories. Each authors had their own classification of PV cooling method. There are three main branches: passive cooling, active cooling, and hybrid cooling. The first type, called passive cooling is divided to: (a) gas/air-based; (b) liquid-based; (c) PCM-based. The second type, active cooling is divided to: (a) gas (air)-based and (b) liquid-based active cooling approaches. The third type, hybrid cooling methods are classified to: (a) Integrating gas and liquid elements; (b) PCM and gas based; (c) PCM and liquid based; (d) Thermoelectric and liquid based. However, for cooling, the most common materials include air, water, PCM, nanofluid, and poly-fluid/material. Based on Ghadikolaei et al. [164], there are two categories of PV cell cooling technology: active cooling and passive cooling. Electric power is not needed for a passive cooling system to carry out its intended cooling of photovoltaic panels. The coolant's natural circulation removes heat from PV. Heat is taken up by cells from their surface and released into the surrounding air. In contrast, active cooling systems rely on external or internal electricity generation to run fans or pumps that remove heat from the surface of the cells. Active cooling has great capability since it uses active components [29]. The cooling system is subject to certain design considerations including load profile, resources, system sizing, PV type and working fluid [13].

3. Passive cooling

One of the efficient methods for cooling the PV modules without using additional energy is passive cooling. This process uses radiation, non-forced convection, evaporation, and spectrum splitting to release the stored heat from a PV module into the surrounding air with the least amount of detrimental effects on the environment. For the purpose of cooling solar farm panels, these passive techniques would be useful and efficient. Every method has advantages and disadvantages in terms of original investment, maintenance expenses, and electricity improvement. According to Siecker et al. [30], in addition to developing methods like liquid immersion cooling, PCM cooling, microporous evaporative foils, and Peltier-based thermo-electric cooling, passive cooling materials include cotton wicks, eutectics, and paraffin wax [165]. The main branches of the passive cooling in this review article will be focusing on air based passive cooling (heat sink/fins), solid-based passive cooling (PCM), and thermal-based (TEG and Thermosyphon).

3.1. Heat sink (fins)

One of the promising passive cooling methods is heat sinks (fins). It can be said that the majority of researchers employ heat sinks made of aluminum to transmit heat. Theoretically, aluminum metal has excellent thermal conductivity, inexpensive, light, and simple to produce [31]. The air circulation at the rear PV module is regulated by fins that increase the heat transfer to cool down the offset heat in the PV. Fins are a crucial part of many systems and gadgets used to enhance air or fluid flow. They are frequently utilized in applications like aerodynamics, propulsion, and cooling systems (Shen et al., 2023).

Three considerations why researchers put attention in the application of this passive-cooling method are because of heat dissipation, improvement in propulsion, and aerodynamic efficiency. Fins are frequently utilized to enlarge the surface area that is ready for natural convection process, which improves heat dissipation. This method is presents in many cooling systems, including air conditioners, electronics, and radiators in automobiles [166]. Besides that, it is a useful instrument in fluid dynamics to improve propulsion. The arranged metal sheets, for instance, are used to improve stability and maneuverability by giving directional control to waterborne vehicles such as ships, submarines, and torpedoes [167]. Lastly, this tool is essential to aerodynamic efficiency, especially in both the automotive and aviation

sectors. Fins can decrease drag, increase stability, and boost overall efficiency by adjusting the airflow surrounding an object.

Depending on the intended use and characteristics, several materials are chosen for fins. According to [32], metals with high heat conductivity, such as copper and aluminum, are commonly used in various applications. Conversely, weight, strength, and thermal qualities can all be balanced with composite materials, for example carbon fiber reinforced polymers (CFRP) [168]. Weight and cost considerations are important in applications where plastics, such as polyethylene and polypropylene thermoplastics, are a good choice for fins [33].

In order to achieve the high heat sink performance, a number of factors should be considered, including surface area, geometry, fluid, and maintenance. (1) Surface Area: Larger surface areas of fins provide better aerodynamic or heat transmission properties [169]. (2) Geometry: Shape, thickness, and spacing of installed fins have a big influence on how well it performs [34]. (3) Fluid: The properties and flow rate of the fluid determine how effective this instrument is. Elevated flow rates typically improve aerodynamic efficiency or heat transfer [35]. (4) Maintenance: Over time, fins may gather dirt, debris, or other impurities that could lower their effectiveness. According to Colombo [36], good operation requires routine maintenance and cleaning.

Several studies have done the investigations using fins as a PV cooling method. Using a numerical simulation, Ahmad et al. [28] determined the thermal performance of truncated multi-level fin heat sinks (MLFHS) for passive PV cooling under non-forced heat release process. The following key conclusions are drawn from the numerical data. (1) The truncated MLFHS performs better in heat exchange than the rectangular plate heat-sink due to its improved surface shape. The suggested form produces an abrupt transition within the boundary layer in the fin restricted region, and better flow patterns are displayed in the design. (2) The reduced MLFHS design yields an average temperature that is approximately 6.13 % lower than the previous version. (3) There is a 2.87 % improvement in the PV module's electrical output power when the MLFHS is shortened. Furthermore, at solar irradiation levels greater than 900 W/m^2, there is a noticeable impact (8.23 %) on the average base temperature decrease. (4) Three-dimensional flow patterns and temperature contours were produced by the numerical study, offering insight into the convection phenomenon that could improve fin design optimization and guide design decisions [28].

In this case, a PV module's fin was positioned behind the panel and operated in a tropical climate with a 15° inclination angle. Fig. 2 displays the heat sink layouts that are the subject of the inquiry. The goal of the proposed shortened MLFHS was to increase the rate of heat dissipation per unit mass. There are ideas that suggest non-uniform fin height heat sinks outperform conventional fin height heat sinks. In addition to the previously mentioned design, the shortened geometry is proposed to generate turbulence, enhancing its ability to transfer heat. Therefore, more research examines the relationship between these aspects is merely suggested [28]. Their findings also indicate that, the MLFHS design produced an average temperature that was 6.13 % lower than the conventional rectangular design. While the previous study focused on fin shape, Marinic-Kragic et al. [37] proposed changing the PV design by making a slit on a PV panel. Their efficient adjustment has resulted in a 3 ° C lower in panel temperature Ahmad et al. [28]; Marinic-Kragic et al. [37].

Elbreki et al. [38] also investigated the PV cooling with fins and a planar reflector through an experimental investigation carried out at the National University of Malaysia. Two different heat sink designs, including longitudinal and lapping fins, were chosen as the passive cooling technique. To determine which cooling system has the lowest payback period when compared to the reference module, economic research has also been done. The later investigation employed lapping fins for passive cooling in an environment with an average solar irradiation of 1000 W/m^2 and a temperature of 33 °C. A mean temperature of 24.6 °C, electrical efficiency of 10.68 %, and power output of 37.1 W were found to yield the greatest results for PV modules. According to the study, in terms of economics, the payback periods for PV modules with longitudinal fins, lapping fins, and reference PV modules are, respectively, 4.2, 5, and 8.4 years. Thus, keeping the PV module's temperature down could be accomplished via a cooling system that has lapped fins. Fig. 3 shows the field study in this experiment [38].

Another fin cooling research done by Hernandez-Perez et al. [39] who investigated aluminum fin with two alternative orientations, one with an average straight fin and the other with segments that are inclined, as depicted in Fig. 4. Using Computer Fluid Dynamic (CFD) software, the heatsinks were first assessed under typical operating settings in order to identify the heatsink with the highest result. It was discovered that segmented heatsinks performed less pressure because they increased turbulence and decreased velocity losses, which improved convective heat transfer. The highest temperature decrease was approximately 9.4 ° C and 10 ° C (for segmented and conventional design), with a conversion efficiency gain of 4 % during 800–1000 W/m^2. Both analytical simulation and experimental findings demonstrated similar result. The findings (Fig. 4) indicate that a 6 W

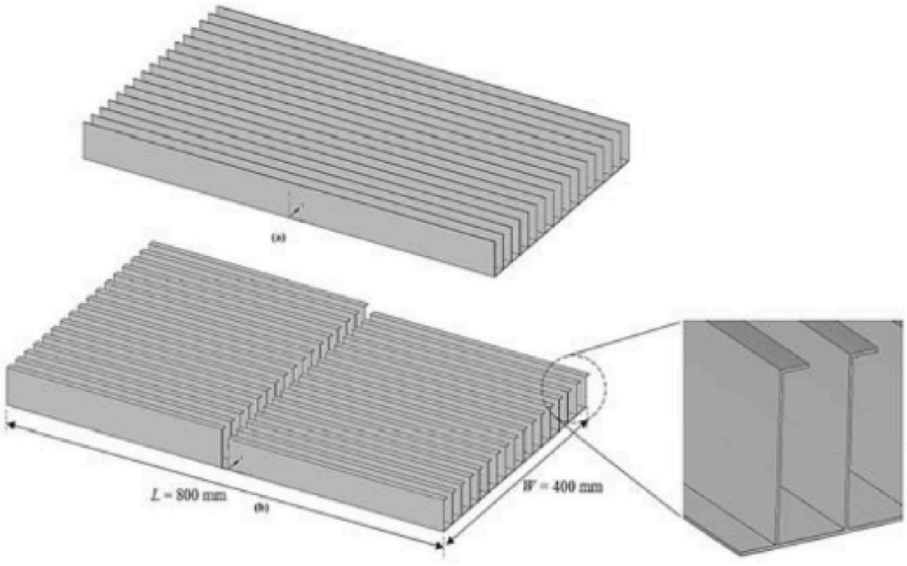


Fig. 2. The geometrical model of MLFHS [28].

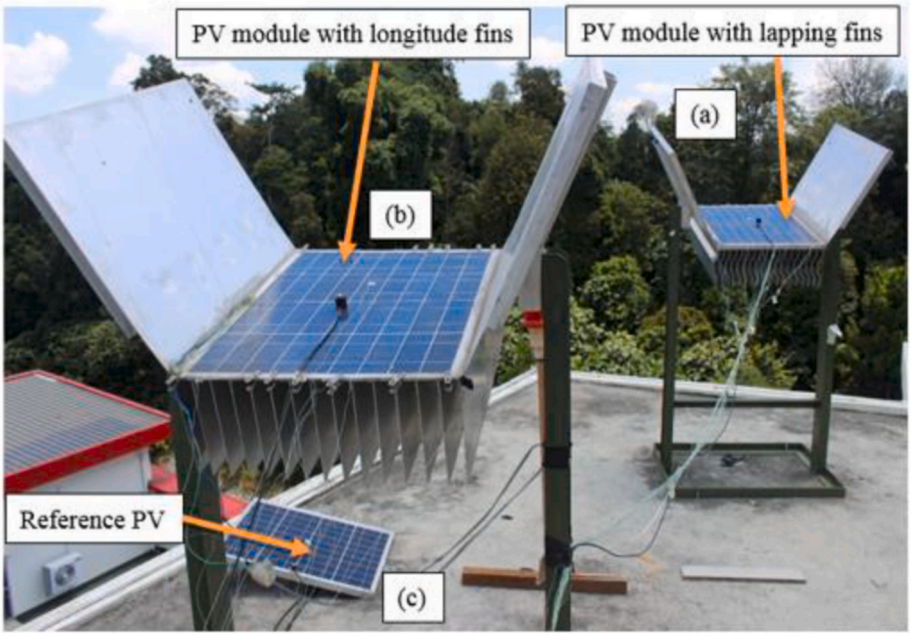


Fig. 3. The field study of the experiment [38].

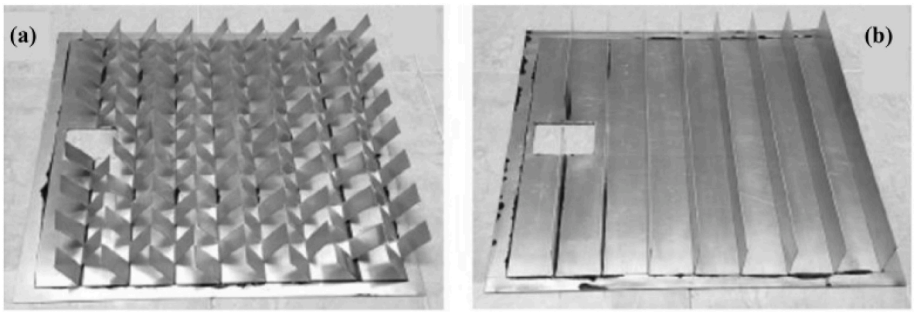


Fig. 4. Hardware model of (a) segmented fin and (b) conventional fin [39].

improvement in capacity per square meter is possible [39].

Furthermore, Abbas and Wang looked into the impact of installing displacement between fins to enhance heat transfer both mathematically and empirically (Fig. 5). The important discovery demonstrates that the displacement reduces heat by as much as 56 %. A decrease in the fin temperature in the upper part affects the reduction in the heat transfer coefficient. Chu et al. investigated the heat transmission capabilities of radiation and natural convection for the triangular fins in an alternating configuration. The heat transfer coefficient will be improved by up to 14.6 % with the proposed triangular fin design and a higher effective view factor. A well-designed triangular fin facilitates easy air entrainment into the fin channel, hence enhancing the heat sink's thermal performance. When triangular fins are positioned vertically, their influence becomes considerably more significant. Altun and Ziylar experimentally examined a wavy-fin heat sink's capacity for convection and radiation heat transfer using a sinusoidal design with variable amplitude. It was found that the rectangular fin design is less effective in transferring heat than the wavy-fin shape. Analogous research has also been done through numerical methods. The suggested ripple fin design offers better cooling performance with a lower base temperature than the rectangular finned plate, which has a mass reduction of 44 %. Due to alleged production issues, this profile shape is not very interesting [40, 170,171]. Table 1 summarizes and compares previous studies that use PV fin cooling.

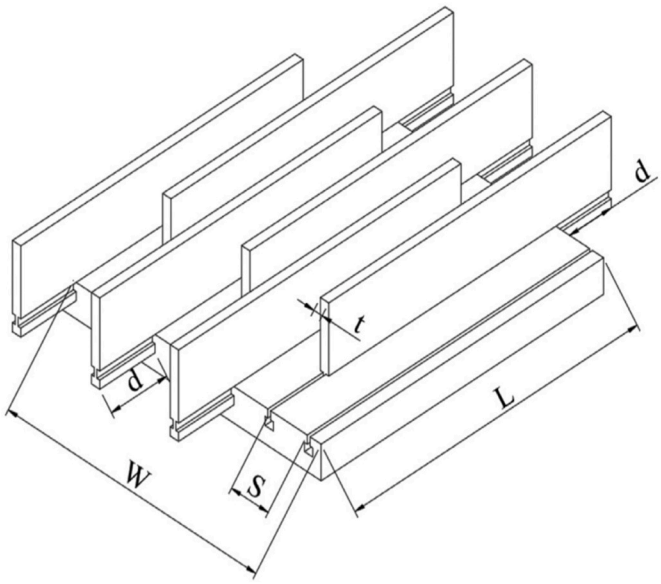


Fig. 5. Heat sink geometrical parameters [172].

Table 1
A comparison between previous studies using PV fin cooling.

Reference	PV type	Cooling method specification	Temperature reduction	Electric performance
Selimefendigil et al. [41]	Polycrystalline PV (75 W)	Closed-cell porous aluminum foam with porosity (75–95 %). Two different fin thicknesses (6 and 10 cm).	Temperature decreased by 49 °C, and 48 °C accordingly.	Power enhancement reached 7.26 W
Grubisic-Cabo et al. [42]	Polycrystalline PV (50 W)	Aluminum fins on the rear of PV module.	When suggested method was tested on a single PV panel, it was shown to have a relative average efficiency boost of almost 2 % during peak power when compared to the reference panel.	electrical efficiency increased as much as 2 %.
Firoozzadeh et al. [43]	Polycrystalline PV (60 W)	Fin cooling	The temperature reduction by 7.4 °*C* and 5 °*C*.	It was determined that there was a 1.328 kW power difference, or around a 20.22 % difference between the modes with and without fins.
Bayrak et al. [12]	Polycrystalline PV (75 W)	Fin air cooling. The aluminum fins were varied to 10 different set, namely by A1-A10.	Temperature reduction for A5 – 45.42 °*C* and A10 – 62.52 °*C* (lowest and highest).	The fins staggered array design yielded the best results, with dimensions of 7 cm by 20 cm. The finned panels (A5) were the highest energy efficiency values at 10.91 %.
Gallero et al. [44]	Low Concentration Photovoltaics (LCPV) (Multi-junction)	Aluminum flat plate base of the fin (20 mm) and the height of the fin (10 mm)	Reduced temperature by 8 °C, Cooling T of up to 32 °C.	The study only focused on temperature reduction only.
Luo et al. [45]	Concentrated Photovoltaic (CPV) (Multi-junction)	Passive cooling with flared-fin.	Cooling temperature of up to 40 °*C* about overall thermal resistance was 10 % lower.	The study was focusing on heat dissipation only.
Amr et al. [46]	Crystalline Silicon Panel	Fin cooling.	The result was about a 4–5 °C for temperature reduction.	The output power recorded 182.7 W and 175 W (with and without fins).
Vittorini et al. [173]	Monocrystalline PV (180 W)	fin cooling with aluminum-based.	Cooling temperature drop by 3 °C.	Gain 2.6 % in electrical improvement.
Hernandez- Perez et al. [174]	Polycrystalline PV (15 W)	L-shape aluminum fin.	Temperature decreased by 10 °*C*.	Improving electrical performance about 4 %.
Elbreki et al. [47]	Crystalline Silicon Panel (40 W)	Lapping fin with reflectors.	Lowering PV temperature by 24.5 °C.	electrical improvement performance by 13.7 % as well as 37.1 W.
Ahmad et al. [28]	Crystalline Silicon Panel	Trapezoidal channel truncated multi-level fin heat sink (MLFHS)	The lowest cooling temperature is 65.4 °*C*.	The electrical output power indicated an improvement of 2.87 %.
Johnston et al. [48]	Crystalline Silicon Panel	Variation in fin heights and inclination angles.	The front and back surfaces of the module decreased by 17 °*C* and 21 °*C*, respectively.	PV module efficiency upgraded by 11.34 % over the reference PV.

3.2. Phase Change Materials (PCMs)

PCMs have a bright future in solar thermal energy applications, and solar storage systems make extensive use of these materials. The majority of PCMs have a high energy density and an unsatisfactory thermal conductivity. In thermal storage applications, methods for improving heat transfer would be taken into consideration in order to lessen the disadvantage of low thermal conductivity. To enhance thermal performance, including nanoparticles into the basic PCM is one suitable method. The latent heat value will drop when the majority of nanoparticles are mixed with a PCM [175]. PCMs can be categorized as organic and inorganic chemical substances. Organic PCMs that are hydrocarbons or fatty acids, such as polyethylene glycol or paraffin wax, are the most prevalent. Inorganic PCMs include salts like sodium sulphate and hydrated salts like calcium chloride hexa-hydrate. PCM can be used in a variety of situations due to its various latent heat capacities, melting and solidification temperatures, and other characteristics [49]. PCMs have the capacity to undergo phase transitions while absorbing or releasing large amounts of latent heat. Usually, these phase changes go from a solid to a liquid or the other way around. These materials' unique capacity to both store and release thermal energy makes them valuable for a wide range of applications [50].

Utilizing PCM to cool PV modules is one of the passive cooling techniques. The output power and lifetime of the modules may be shortened by excessive heat. The PCMs materials can lessen these problems and enhance the efficiency of PV systems [22]. This method plays the role as a thermal energy storage medium when coupled with PV modules. It absorbs heat that is not needed from the module during times of intense solar radiation or ambient temperature. It turns from a solid to a liquid as a result of this. Excess thermal energy is efficiently stored by this process inside the PCM. When it solidifies once more, releasing heat that has been stored back into the environment as the outside temperature drops or solar radiation drops. PV module temperature is controlled by this heat transfer method, which also guards against overheating [176].

PCMs have a high latent heat capacity, making them suitable for large-scale heat storage and useful as coolants for PV cells. Underneath the cells are channels filled with PCM. By raising the temperature of the PV along the sun shining period, the PCM absorbs the heat from PV; at night, it releases the heat into the surrounding air and solidifies as illustrated by Fig. 6. The primary drawback to PCM cooling is the comparatively high price of raw sources; otherwise, PCMs work best for cooling PV cells in regions with minimal seasonal temperature variations [164]. PCM integration improves temperature management in photovoltaic systems, which raises output and boosts dependability. Additionally, PCMs may minimize demand on active cooling methods like liquid cooling and air cooling with pump or fans respectively, which lowers maintenance needs and energy usage. As consideration, PCMs present a viable means of cooling PV modules through effective thermal energy management. They are a useful passive cooling technique that improves the efficiency and lifespan of solar systems due to their capacity to absorb and distribute heat during material changing [51].

The following are some advantages and disadvantages of PCM cooling technique:

The advantages are.

- ➢ A notable increase in PV system performance of applying the PCM cooling technique.
- ➢ Making use of the heat that has been preserved due to the PCM cooling mechanism behind the PV surface.

The disadvantages are.

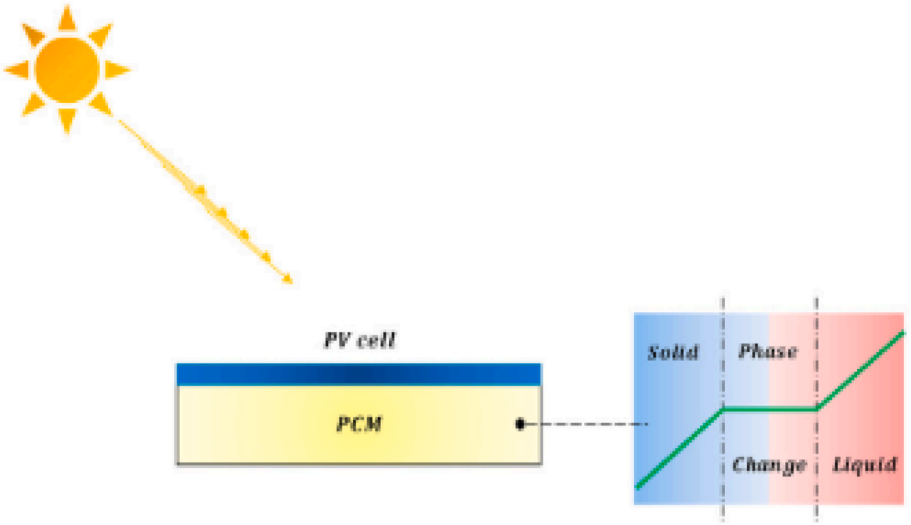


Fig. 6. An illustration of PCM cooling process [164].

- ➢ Selecting the right PCM is entirely dependent on the climate condition; using the wrong PCM might provide different results and harm PV cells.
- ➢ Only areas with a hot environment and minimal seasonal temperature changes are suitable for using the PCM cooling approach.
- ➢ PCM's gradual reduction in heat absorption [52,164].

In order to maximize power generation, Sun et al. conducted research on the application of phase change material (PCM) with the appropriate thickness to control the temperature of the solar cell module. A PCM known as RUBITHERM RT42 was used to absorb heat at the back of a 250 W polycrystalline solar cell. Its melting point was 42 °C, and its thickness was 50 mm. The numerical enthalpy approach was used to forecast the temperature of the solar cell module and the melting processes of PCM, RT35, 42, 47, and 55. The outcomes and the experimental data agreed fairly well. The maximum power output was found to be about 167 W at 867 W/m^2 of solar light. The daily electrical outputs that corresponded were 0.642 and 0.707 kWh. The simulation carried out in the Chiang Mai environment revealed that RT42 PCM should have a thickness of about 40 mm [53].

Kumar et al. carried out a numerical analysis of the PCM container shape and its effect on the RT 27 PCM melting rate. The rectangular container that was utilized as a reference was contrasted with three non-rectangular shapes. The PV temperature was lowered and electrical efficiency was increased by each of those linear, parabolic, and cubic designs' superior PCM melting capabilities over those of conventional rectangular designs. In comparison to the other two PCM enclosures, the cubic enclosure has superior performance. Fig. 7 (a) and (b) show the system schematically with a conventional PCM enclosure and a customized PCM enclosure, respectively. There are still certain gaps in their research report that prevent their findings from being applied to a real-world scenario because it was a numerical simulation run vertically and with constant solar radiation, rather than taking different tilt degrees into account. Therefore, it is advised to conduct an outdoor experimental investigation in this field [54].

As an illustration, Elsheniti et al. [55] developed a one-dimensional

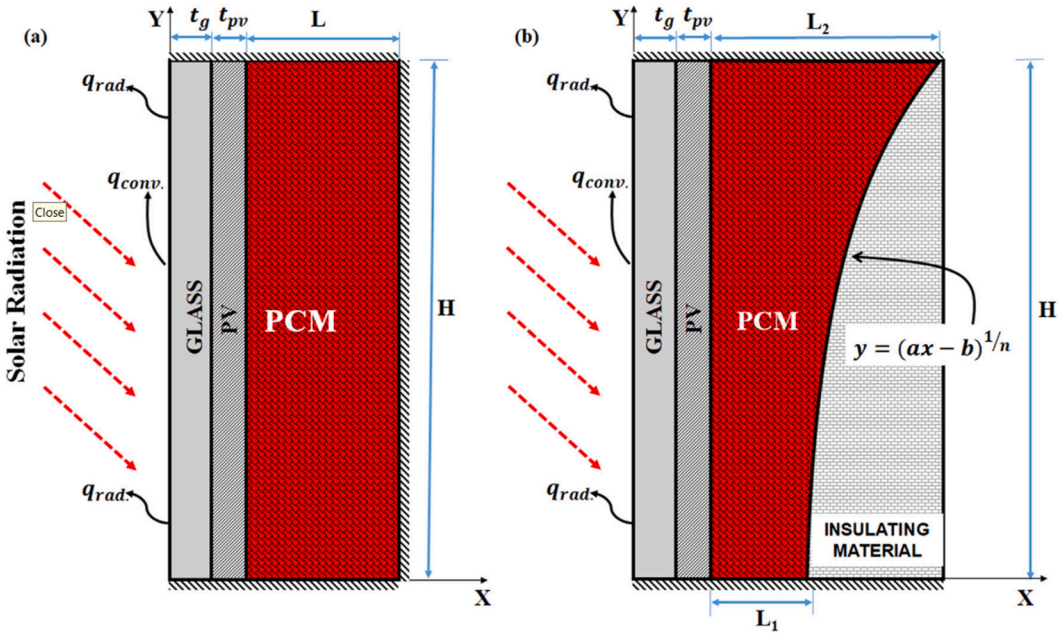


Fig. 7. (a) Illustration of cross-section of PV with PCM; (b) modified standard container [56].

mathematical formula to predict the temperature of PV in contact with RT 27 PCM. They changed the PV module orientation from 0° to 90° using a numerical method and applied a variable solar flux based on the location of Alexandria, an Egyptian city, at different times. Their abridged mathematical model demonstrated a plausible inconsistency with reported experimental findings. They were therefore successful in cutting down on computation time. According to their research, the ideal inclination angles for four different seasons were determined to be 18°–48°, and 41°, respectively. Additionally, during the high-incidence period, various PCM thicknesses of 10, 20, 30, and 40 mm were investigated. The PCM with a thickness of 30 mm produced the greatest rate of 18 % average efficiency, according to the results. Higher depths were unable to consolidate during the day and night whereas lower depths evaporated quickly [55].

Comprehensive laboratory experiments on several silicon solar cell cooling techniques have been conducted by Ahmadi et al. [57]. In their study, they took into account the concepts of passive, active, and their combinations. However, only passive techniques are covered here. The experimental setups for passive cooling techniques are displayed in Fig. 8. The carbon nanotube foam implantation in the PCM composite increased paraffin's low heat conductivity. The PV module's output voltage and glass temperature were assessed between both sets of results. Clearly, using PCM-composite will keep the temperature less for an extended amount of time, and using PCM may help the module stay cool under continuous radiation for an extended amount of time. It is anticipated that maintaining lower temperatures will prevent voltage drop [57].

There are various methods to enhance PCM's ability to maintain PV modules at lower temperatures. It has been suggested that nanotechnology can be used to improve PCM's heat transfer capabilities [58,59]. The temperature of a PV was regulated using RT55 paraffin wax in an examination by Nada et al. [60]. The second technique for temperature control was mixing 2 % of alumina nanoparticles with pure PCM. The findings demonstrated that the addition of nanoparticles and the use of PCM in pure condition led to temperature reductions of 10.6 °C and 8.1 °C, respectively, in comparison to the non-cooled PV panel. The aforementioned cooling system's efficiency increased by 5.7 % and 13.2 %, respectively, as a result of these temperature drops. The increase in thermal conductivity was linked to the improvement in heat transition impacted by the use of nanoparticles in PCM [58–60]. Table 2 illustrates the recent PCM cooling applications in PV.

3.3. Thermosyphon

It is suggested that an active cooling channel during the day be replaced with a passive radiative cooling system to boost the efficiency of a typical PV module. To capture cold energy, water can be pumped beneath the photovoltaic panel, and the module itself functions as a radiative cooling surface at night. Yoon et al. [178] demonstrated the possibility of night-time, power-free water circulation in a closed-loop using the thermosyphon effect. But, throughout the day, by passing cold water that has been stored through the water channel, the PV module can be cooled [178]. In order to simplify the experiment setup, a PV module measuring one by 1 m squared is chosen. The nighttime

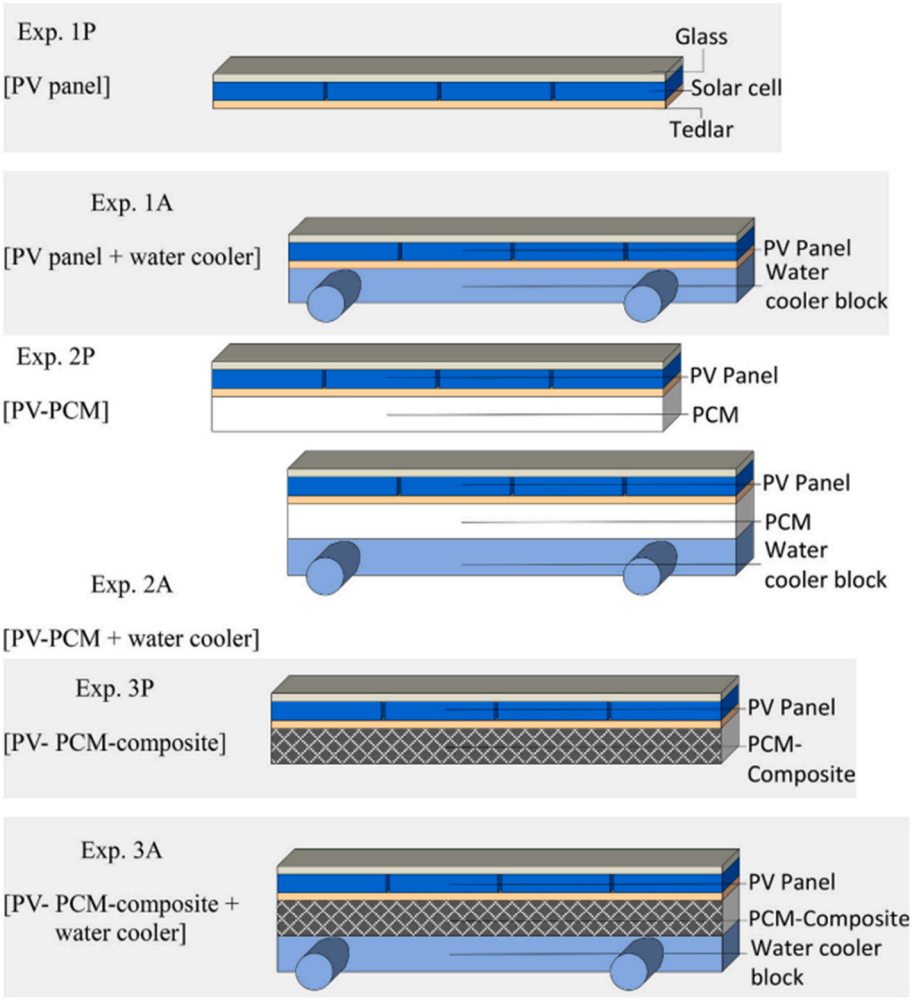


Fig. 8. Passive experimental configurations [57].

Table 2
Recent PCM cooling applications in PV.

Reference	PV type	Cooling method specification	Temperature reduction	Electric performance
[61]	Silicon PV module (140 W)	PCM cooling with double skin facade	PV temperature lowered by 10 °C, 20 °C and 17 °C	Enhancing power output by 5 %, 8 % and 5 % respectively.
[62]	Crystalline Silicon Panel (10 W)	PCM box attached at behind Paraffin	The module with PCM made possible reduction to the lower temperature by 7 °C	Electrical improvement is 7 % larger and rise to 1080 kWh/year.
[63]	Monocrystalline PV module	PCM with copper pipes PCM RT 27 and PCM RT 31.	The temperature was decreased by 6.4 °C and 7.5 °C, in separate PCM design.	Electrical output efficiency jumped by 4.19 %.
[56]	Poly-crystalline panel, 12Wp.	PCM with $CaCo_3$, SiC and Cu material-based.	Temperature went down by 2 °C with average value of 4.4 °C	The average of 2.2 % more electricity efficiency can be achieved by using integrated PCM.
[64]	Mono-crystalline Panel, 175Wp.	PCM – Calcium chloride hexahydrate-based	The first scheme, Temperature was fallen by 13.3 °C and 18.0 °C for first and second trial, respectively.	The result for the electrical efficiency was running twice, first trail pointed at 6.2 % and second trail pointed at 8.3 %.
[65]	Crystalline Silicon Panel	PCM with various mixture materials.	Eutectic PCM could lower the PV temperature by 10.3 °C.	This method could upgrade electrical efficiency by 10.09 %.
[66]	Mono-crystalline PV	Fusion in PCM and Calcium chloride hexahydrate	The maximum temperature reduction was 26.3 °C	24.68 % of electrical output improvement.
[67]	Poly-crystalline PV (20 W)	PCM OM 35 cooling.	Temperature reduced by 32 °C	PCM cooling could lift efficiency in electrical performance by 4.5–8.5 %
[68]	Poly-crystalline Panel, 10Wp.	PCM – Composed by 82 % coconut oil and 18 % sunflower oil.	Temperature reduction by 20 °C–32 °C and 20 °C–29 °C.	Electrical Efficiency improvement for composed oil - 21.1 %, 26.8 %; with Composed oil 29.24; and Nano PCM - 44.7 %, 46.6 % and 48.2 %, respectively.
[69]	Mono-crystalline	PCM with Graphene	Temperature reduction is	Electrical efficiency
[177]	Crystalline panel, 310Wp	Pure PCM paraffin wax, and PCM Paraffin-expanded graphite	Average temperature reduction is 6.7 °C	Electrical efficiency improvement: at 2.5 cm pPCM by 14.29 % and cPCM 14.7 %; At 5 cm thickness, pPCM 14.1 % and cPCM 14.86 %.
[60]	Poly-crystalline Panel, 30Wp.	PCM - RT55 and NPCM - Al_2O_3 nanoparticles were used.	Temperature was reduced by 10.6 °C and 8.1 °C for NPCM and PV-PCM.	The electrical efficiencies were 13.2 % and 5.7 % for PV-NPCM for PV-PCM, respectively.
[70]	Crystalline Silicon Panel	PCM cooling	Temperature had been reduced by 24.9 °C	The output power had been increased by up to 11.2 %.

thermosyphon effect is produced by placing the storage tank underneath the PV module. By installing the PV module on the top of the reservoir, it is possible to passively circulate and collect the water that is frozen via a channel beneath the cold surface of the PV module at night. The stored cold water may be utilized as an operation fluid that cools the PV module during the entire day using a mechanical pump. Because of this, during the day, the water channel beneath the PV module acts as an active cooling system as can be seen in Fig. 9 [179].

The temperature inversion hampered the improvement in power generation, especially in a small storage tank. It was also found that, as the distribution of heat through the reservoir may offer an additional pathway for heat dissipation, it is advisable to keep the tank uninsulated. It was shown that there was a considerable correlation between the efficiency gain and climate, namely the diurnal fluctuations in temperature. In particular, the average yearly diurnal temperatures in Phoenix, Dallas, Miami, and Los Angeles are 13.7, 11.8, 7.3, and 7.0 K, respectively. The annual total of net power percentage might increase by 6.4, 3.3, 2.5, and 1.8 %. Therefore, it can be said that the suggested method can function well in areas with significant yearly variations in daytime temperatures, which are typically found in hot, dry regions. The suggested RC-assisted PV module system may provide an alternative method of cooling solar modules for effective power production [178].

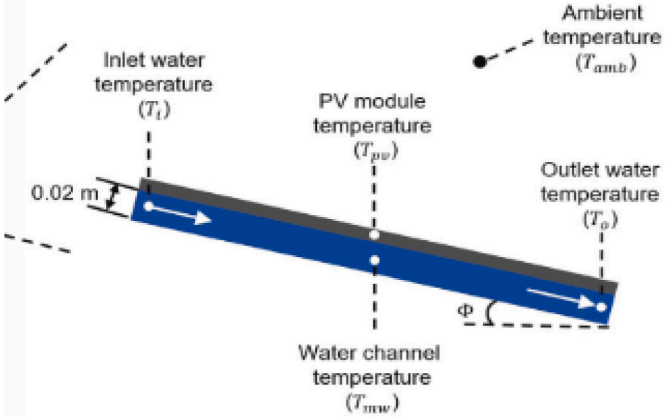


Fig. 9. Schematic of the experiment [178].

In their work, Wang and the lab team have designed a new kind of atmospheric plate thermosyphon (APT) cooling system that can be used for heat dissipation from single solar cells or low concentrations of material. The atmospheric plate thermosyphon (APT), a novel form of cooling device that can be used to remove heat from single or low-concentration solar cells, is suggested. The temperature performance and heat transfer performance of the evaporator were the main topics of inquiry. The average PV temperature increases from 850 W/m^2 to 3200 W/m^2 at a 60° tilt angle, although not by much above the boiling point temperature of the working fluid, which varies from 0.16 to 4.1 °C. The condensate temperature does not simultaneously alter the evaporator's average temperature by more than 1 °C [71].

3.4. Thermoelectric generator (TEG)

The thermoelectric generator (TEG) system can operate as a cooling system when voltage is used to the p and n sides of the system—a process known as the Peltier effect. On the other hand, the Seebeck effect can produce electricity if there is a difference in temperature between the two legs. Regarding the Peltier effect, TEG is categorized as active cooling since it can use an amount of the power output to cool a solar module. However, if the Seebeck effect occurs, the TEG may be able to supply more power by capturing the excess heat from the PV module. As a result, the PV module cools down and produces additional power simultaneously. It would be advantageous to view this technology as hybrid [8]. Wu et al. [180] have studied these two-hybrid systems in a variety of settings since the TEG system coupled with PV cell can have two separate states with reference to the Peltier or Seebeck effect. Reports state that although the PV-TCS (hybrid PV with thermoelectric cooling) has a smaller temperature of the cells than the PV-TGS, it has less valuable power and efficiency [180].

The Peltier and Seebeck phenomena can be employed as suitable countermeasures to improve the usefulness of solar systems in different designs, as has already been noted quite a bit. The Peltier effect is typically maximized or waste heat from the solar system is recovered through the Seebeck phenomenon by using thermoelectric thermal management modules, which increases the solar system's efficiency. A number of recent review articles have addressed the application of thermoelectric apparatus in solar systems. The use of thermoelectric modules to enhance solar desalination systems, the use of TEGs to generate electricity from solar ponds, the use of TEGs to generate power in solar thermal systems, and PV-TEG hybrid systems are a few of these topics covered [72,73].

Through the use of TEG and thermoelectric cooling, Yang et al. [181] enhanced the PV module's electrical power productivity. Solar energy was collected on the PV panel using a concentrator. The thermoelectric components, which functioned as an energy converter, was placed between the heat sink and PV panel. It is composed of several conducting, ceramic, and p-n thermocouples plates. Through a number of cooling methods that employ either water or air as a cooling medium, the thermoelectric module's missing end released waste heat. Two variations of the concentrated PV thermoelectric (CPV-TEG) systems were evaluated with reference to this figure: the concentrator PV thermoelectric cooling arrangement and the CPV-TEG system. Investigators found that the exergy-economic analysis from this proposed system were equivalent to 0.266 and 0.263 $/kWh, respectively, at a solar intensity of 1000 W/m^2. According to the results, a water nanofluid and TEG combination produced the maximum power generation in a photovoltaic system utilizing nanofluid, which was approximately 47.7 % more than a traditional system. Additionally, the usage of TEG and PCM in Ardabil, Iran, improved the PV system's electrical performance and power generation by 10 % and 1.38 %, respectively. Moreover, the performance of the PV system was improved by thermoelectrically cooling the solar panels [181].

In order to decrease the outside temperature of the PV top layer and increase the efficiency and electrical output, Soltani et al. [74] implemented TEG and cooling techniques. The PV panel was coupled with TEG as an effective means of capturing waste heat and utilizing it to generate electricity. The cooling technique used on the thermoelectric generator's cold side has a significant impact on both the obtained power and electrical efficiency. Five distinct cooling fluids—natural, air, water, SiO2 nanofluid - water, and Fe3O4 nanofluid -water—were the subjects of experimental study. Researchers found that SiO2-water cooling provided the best electrical power and efficiency when compared to the natural cooling approach; these values were raised by 54.29 % and 3.35 %, respectively. Fig. 10 displays the PV panel attached to the TEG that uses cooling techniques [74].

Lekbir et al. [75] examined how different cooling method configurations affected the performance of the PV module. The PV module is connected to the hot surface of the TEG in order to use the waste heat from the panel. The TEG's cold side temperature is lowered by the water-carbon nanofluid that flows underneath it. The PV panel had a one square meter area and a mass flow rate of about 0.0104 kg/s for its coolant, which was a carbon nanotube-water nanofluid. According to their findings, the PV panel-TEG using nanofluid had an electrical performance that was 49.5 % better than that of normal PV panels [75]. In another situation, Dimri et al. evaluated the impact of air flow and TEG on the electrical performance of the PV module. The experimental setup was evaluated in three different configurations: one with an air pipe, one with a PV panel partially enclosed with TEG, and one with a PV panel fully enclosed. An opaque PV panel incorporates 36 TEGs as part of the collector structure. The TEG has measurements of 4 × 4 × 0.04 cm. They found that PV panels with full covering have a total electrical efficiency that is 4.46–6.23 % greater than PV panels with partial coverage. PV utilizing TEG is depicted in Fig. 11 [76].

Another experiment by Dimri et al. [77] in different year examined three distinct system scenarios in the evaluation of various PV thermal thermoelectric systems: an opaque PV thermal thermoelectric air and water collector, an aluminum PV thermal thermoelectric water collector, and a semitransparent PV thermal thermoelectric water collector. Thermal modelling and artificial neural network modelling were taken into consideration in order to assess the PV module's electrical power generation. The hot side of the TEG was positioned below the PV thermal. The general electrical and thermal efficiency of opaque PV/T systems were improved by 2.8 % and 21.8 %, respectively, with the application of thermoelectric module water collecting systems. Additionally, they demonstrated that the results of the artificial neural network model and the thermal model agreed well. The various PV thermoelectric system instances are depicted in Fig. 12 [77]. Table 3 summarizes the work done on several TEG and Thermosyphon PV cooling methods.

4. Active cooling

PV panels generate electricity by converting sunlight into electricity through the photovoltaic effect. Nevertheless, the efficiency of the PV panels tends to decline with increasing temperature. PV active cooling systems aim to mitigate this efficiency loss by actively cooling the solar panels, thus maintaining their temperature within an optimal range of performance. Pumps, fans, or blowers are needed in active cooling solutions to flow the coolants (water, air, or nanofluids) [88].

4.1. Water cooling

Water cooling in general is an active cooling method which using water as coolant and circulate it by using pump, sprayer, pipe, and etc. One of the water-cooling methods to enhance solar systems is by using spray nozzle cooling. By using focused water sprays to effectively dissipate heat from the PV panels, this technique helps to regulate temperature and may even improve performance. The configuration of the system, which includes the nozzle type, tube diameter, stream rate, and irrigation method (which may be automatic, continuous, or

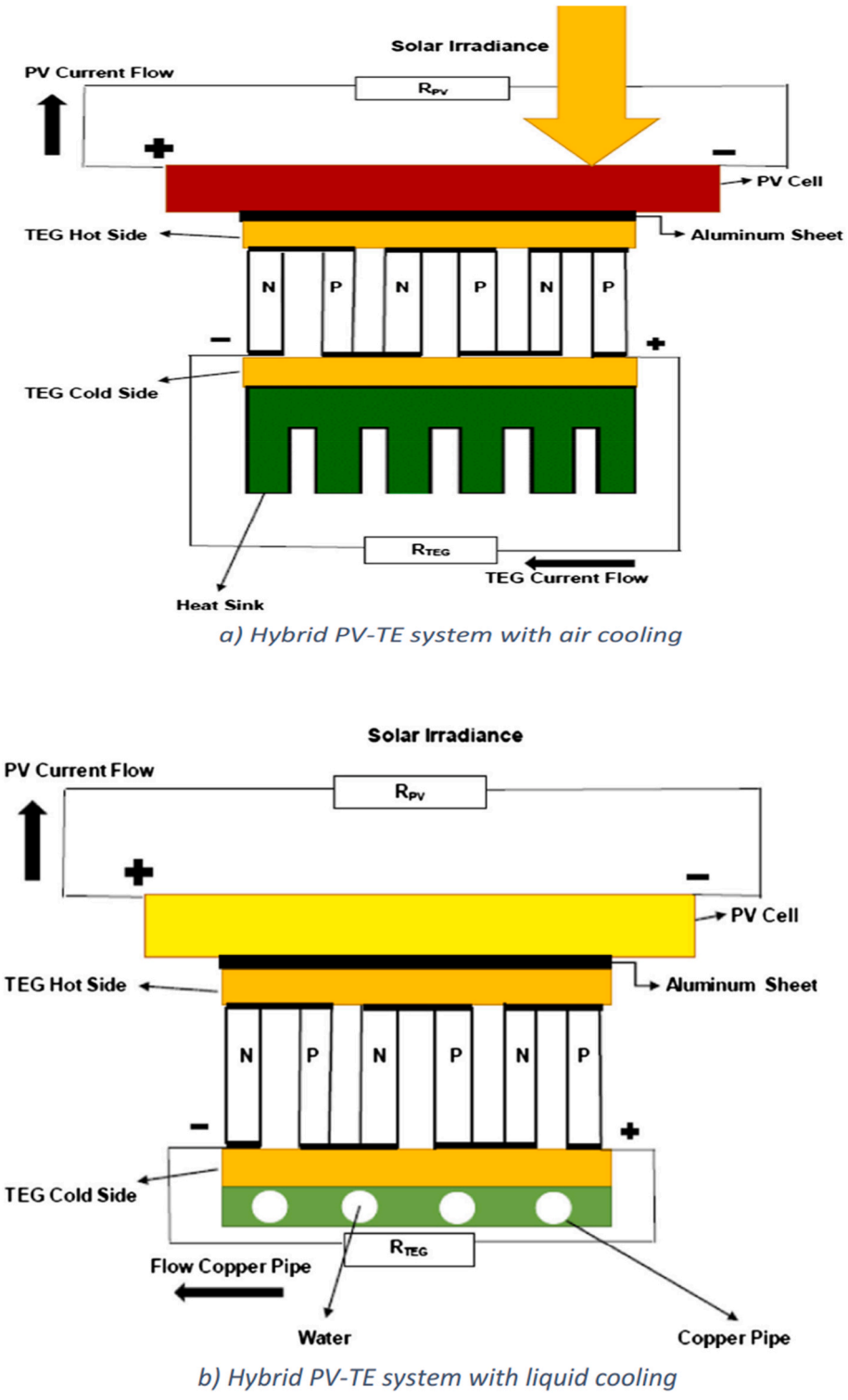


Fig. 10. Diagram view of combined PV with TEG using (a) air and (b) water cooling [74].

pulsating spray), primarily dictates how well a water spray cooling method works. One evaporative cooling technique utilized in active mode is spraying water over solar panels (Fig. 13). Nizetic et al. [89] looked at the effects of a water spray cooling method on the two sides of the PV module during the times of maximum solar radiation. Electrical power and electrical efficiency have grown by 16.3 % and 14.1 %, respectively, according to the data. Furthermore, if the front and rear sections of the PV module are cooled concurrently, the temperature drops from 54 °C (without cooling) to 24 °C. Thus, it may raise the electrical output by more than 17 % and lower the PV temperature by up to 25 °C [89].

The possibility of using high frequency ultrasonic to improve the cooling capability of photovoltaic modules has been explored experimentally by Rostami et al [90]. Atomized CuO nanofluid (0.01–0.8 W/V) and pure water were used as the heat transfer fluids. The cooling efficiency of PV modules has been found to be greatly enhanced by atomizing liquids with ultrasonic vibrations. It was shown by the results that atomized nanofluid cooling is more effective than atomized pure

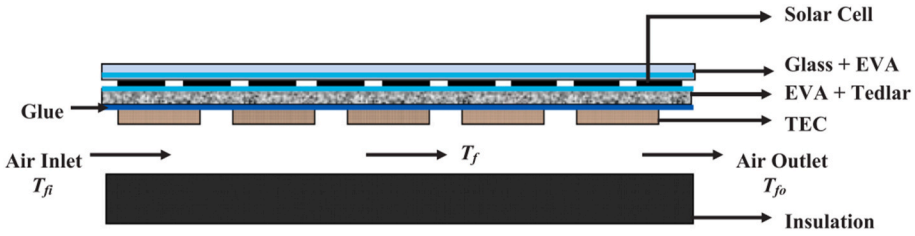


Fig. 11. Sketch of PV panel and TEG module [76].

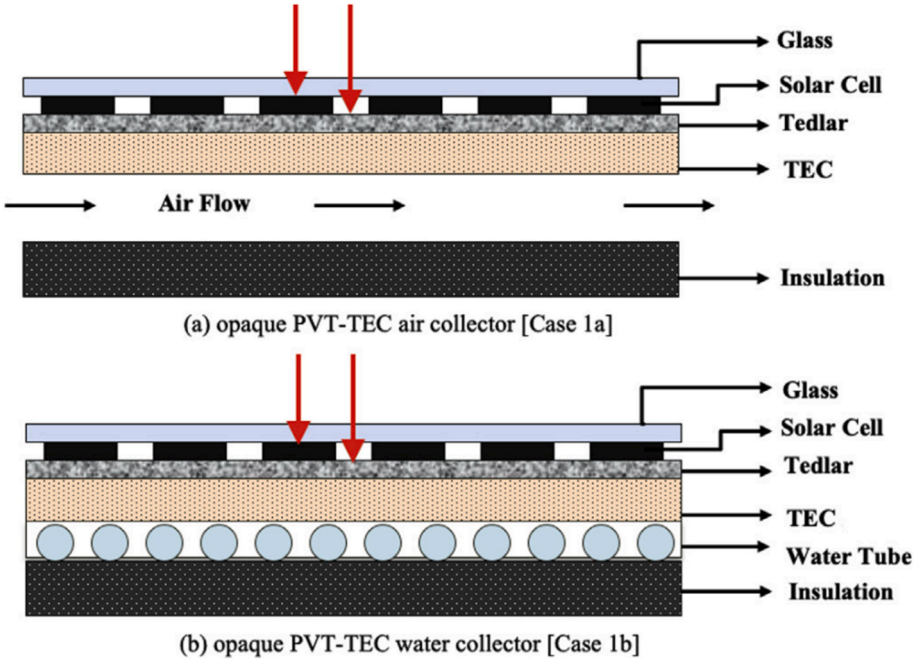


Fig. 12. Schematic of various PVT using TEG [77].

Table 3
Summary of several TEG and Thermosyphon works and results.

Reference	Cooling method specification	Temperature reduction	Electric performance
[78]	PV-TEG cooling.	Temperature reduction is 2.68 °C, 2.53 °C, and 2.35 °C.	More power could be gained by 1.35 % at maximum hybrid concentration.
[79]	Using concentrated PV-TEG hybrid system and CPV-PCM-TEG	The gained significant temperature reduction by 30 °C.	Conducted electrical improvement by 23.52 %
[80]	PV-TEG integrated with PCM variation.	Temperature decreased by 1.5 °C and 3.9 °C	Improved the general electrical effectiveness by 4.52 %.
[81]	PV-TEG	An optimum cooling progress reached overall efficiency of almost 25–30 %.	The increased efficiency was up to 10 %.
[74]	PV-PCM-TEG.	The temperature was fallen almost by 10 °C.	The power production and efficiency were climbed to 54.29 % and 3.35 %, respectively.
[82]	PV-TEG arranged with fin.	The temperature could be reduced as high as 10 °C in average.	Power and efficiency both claimed same average increases of 10.50 %.
[83]	PV-PCM-TEG and water cooling	The temperature reduction was 32 °C.	The PV-PCM-TEG hybrid system with water cooling had a greater efficiency than the one with air cooling, namely 16.70 % and 12.54 % accordingly.
[84]	PVT-PCM-TEG.	The thermal condition was decreased by 1.77 °C, 9.76 °C, and 13.17 °C separately.	Efficiency in electrical aspect improved by 13.09 %, 16.17 %, and 20.68 % separately.
[85]	PVT integrated with thermosyphon	Degradation in temperature number was 16.35 % or 45.14 % (methanol and acetone).	The electrical efficiency raised to 5.67 % and 7.7 %, (methanol and acetone).
[86]	PV-TEG with heat pipe	The average temperature difference is 15–20 °C.	Wind speed and irradiance boosted PV panel efficiency.

Reference	PV Type	Temperature reduction	Electric performance
[87]	Nanofluid in a two-phase closed thermosyphon (TPCT)	Degenerated temperature to 14.52 °C.	Generated 1.42 W power and 1 % average electrical energy performance.
[71]	Atmospheric Plate Thermosyphon (APT)	The highest temperature difference occurred at 5.6 °C after APT started operate.	The study focused on thermal characteristic of the APT cooling System.



Fig. 13. The water spray cooling experimental setup using the designated water nozzle system: a) upper, b) rear [89].

water cooling. Additionally, the maximum results power and cooling function of the PV module are suitably impacted by increasing the density of nanofluids. In comparison to the mode without cooling technology, atomizing 0.8 (W/V) of nanofluid led to a 57.25 % decrease in module temperature and a 51.1 % rise in maximum power. These are the early research results. Fig. 14 displays the design layout for this work, which uses high frequency ultrasound [90].

In the Mah et al. [91] investigation, a rooftop photovoltaic system that had been refurbished and was operating as a distributed generating system had a water-film cooling mechanism installed. The experiment indicates that this approach reduces the thermal tension between the front and back surfaces of the PV panels as well as the non-uniform distribution of temperature among the various PV positions. Additionally, the cooling system with an ideal cooling water flow rate of 6 L/min can raise the generation of power by 32 W per 260 W-rated PV module (15 % growth) at 1150 W/m2 of solar irradiation, resulting in a net energy gain of 0.0178 kWh/hour/panel [91].

Bevilacqua et al. [92] in another view evaluated the long-term effectiveness of three distinct forced ventilation and spray cooling systems applied to the backside surface of PV modules. On days with lots of sunshine, the cooling systems might lower the backside temperature by up to 26.4 °C. The research also confirms that the distribution of temperatures is uniform. The spray cooling system's impressive capabilities were validated by its August efficiency of 14.3 %, which was higher than the reference module's 12.7 %. The monthly rise in electric energy for the system that performed the best ranged from 1.4 % in December to 8.6 % in June. Throughout the course of the investigation, there was the biggest increase in energy yield—6.1 %. The efficiency analysis provided additional support for the findings, as the spray cooling system-maintained conversion efficiency over 14.5 % for the full day in May and October, and near the nominal value in August [92].

In Hadipour research team's work, a water-cooling system with pulsed spray is built for PV to increase their performance and reduce water usage during cooling. The performance of the PV with the pulsed-spray water cooling system is compared to that of the steady-spray water cooling system and the uncooled panel. A cost assessment is also conducted to determine the financial benefit of implementing the new cooling systems for the PV panels. When using steady-spray water cooling or pulsed-spray water cooling, respectively, a solar panel's maximum electrical power output increases by about 33.3 %, 27.7 %, and 25.9 % when compared to an uncooled panel [93].

The majority of studies on the evaporative cooling of solar modules have mostly depended on experimental investigations without a generic model to examine the performance of the module. In order to determine the thermal characteristics of the module, overall heat loss coefficient and optical efficiency, temperature, and generated electrical power, an experimental model identical to that of solar collector performance and was developed by Chea et al. [94]. It measured total efficiency as well as the ratio of temperature difference between ambient temperatures and cooling fluid temperatures with solar radiation. In their work, a PV module having an evaporating surface and water flow at its back surface was subjected to experimental testing. In comparison to the usual unit, the average total and net monthly electrical production rose by 7.04 and 4.47 %, respectively. In a hot and dry region like Ouargla, Algeria, PV module performance might be improved further, leading to an increase in average total and net monthly power output of 8.62 and 6.48 %, respectively. The experimental photoshoot is shown in Fig. 15 [94].

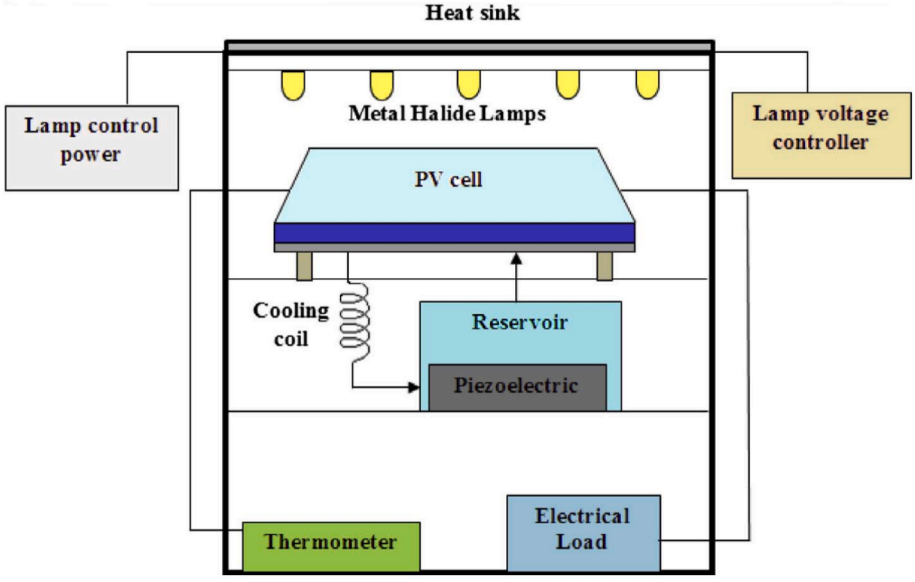


Fig. 14. Schematic illustration of the feasible high-frequency ultrasound experimental setup [90].

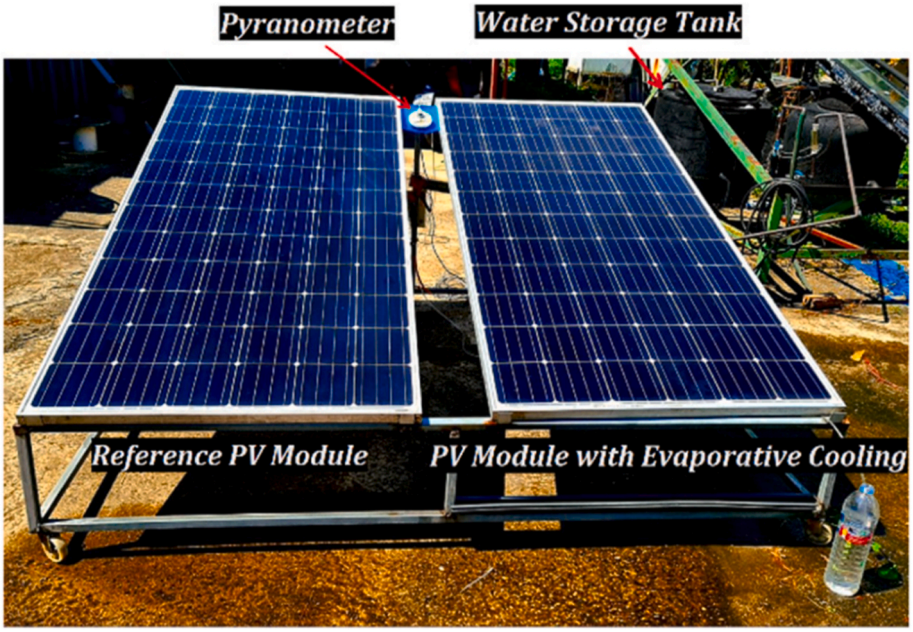


Fig. 15. An experimental photoshoot of cooled and uncooled PV [94].

Table 4
Various active water-cooling investigations.

Reference	PV type	Cooling method specification	Temperature reduction	Electrical performance
Ebaid et al. [95]	Monocrystalline PV (50 W)	Water cooling with various nanofluid mixture.	Temperature decreased to 57 °C	Average enhancement for electric output were 4.42 % and 2.25 %, respectively in term of Voc and Isc.
Alshaheen et al. [96]	Monocrystalline silicon PV (40 W)	Applying nanoparticles in nanofluid cooling.	The temperature reduction in percentage from PV increased by 4.78 %, 4.8 %, 4.73 %, and 4.5 % for different experimental setup	The energy efficiency for each scheme increased by 19.3 %, 15.24 %, and 9.46 %.
[97]	Poly-crystalline PV (100 W)	Water cooling integrated with brand new dual oscillating absorber copper pipe design.	In the indoor experiment and the theoretical model, the highest cell temperature was 57.32 °C and 58.73 °C, respectively.	The maximum electrical performance reached 11.5 %.
Liu et al., 2020	Crystalline Silicon Panel	Water cooling system with various parameters were studied.	The PV surface temperature may be efficiently lowered by the optimized PV cooling system, and it is approximately 47 °C smaller than that of the uncooled PV.	The power generation had growth by 7–15 %
[98]	Polycrystalline PV (10 W)	Water cooling with copper tubes.	The percentage of decreased temperature was by 15.23 %.	The electrical performance climbed by 6.08 %.
[91]	Crystalline silicon PV	Using thin-film for water cooling	The average reduction in PV temperature was 28 °C.	At the best cooling rate, the output power was improved by 15 %.

Reference	PV Type	Cooling method specification	Temperature reduction	Electric performance
Tashtoush and Al-Oqool [99]	Polycrystalline PV (240 W)	Water cooling	Amount of reduced temperature was 37 °C.	Generated extra energy production by 15.28–17.75 %.
Abdallah et al. [100]	Monocrystalline PV (10 W)	Water flow rates are 0.6, 0.9 and 1.2 L/min.	Alumina nanofluid reduced 8.6 °C of the PV module temperature.	Average improvement in electrical efficiency was 24.6 %.
Abdallah et al. [182]	Monocrystalline PV (10 W)	Flow rate is 1.2 L/min.	Using MWCNTs-water based Nanofluid, a temperature reduction of 12 °C for the PV panel.	Average improvement in electrical efficiency was 13.58 %.
Al-Waeli et al. [101]	Polycrystalline PV (120 W)	Water cooling with SiC/ deionized	Nanofluids coolant was able to cool down 28.1 % of temperature of the PV.	Electrical efficiency reached 14.25 %.
[92]	Polycrystalline PV (245 W)	Spray water cooling and forced ventilation cooling.	Both cooling types were capable to lowered rear PV temperature by 26.4 °C and 21.8 °C accordingly.	Obtained the highest electrical performance by 14.3 %.
[93]	Monocrystalline PV (85 W)	Water cooling with pulsed-spray method	The temperature fell from 57.1 °C to 24.8 °C.	The minimum water consumption could increase the output power as much as 27.7 %.
[102]	Polycrystalline PV	Water cooling with automated spray.	The study only focused on electrical properties.	The electrical efficiency of the of PV added by 16.65 %.
[94]	Monocrystalline PV	Evaporative cooling	Temperature lowered 15–23 °C.	Generated power increased over 15 %

Table 4 illustrates several active water-cooling investigations.

4.2. Air cooling

Compared to the natural convection model, forced air cooling gives a higher rate of heat transfer, and the high mass flow rate is going to control and maintain the PV cell temperature at an appropriate level. Since air requires material, it is regarded to be a less expensive alternative medium. Therefore, forced air flow or induced convection are better choices for PV or PVT working fluid [103]. In order to test the efficiency of solar cells with forced air cooling and supported by a heat sink cooling system, Soliman et al. [104] executed an indoor

experiment. In forced and natural air circumstances halogen lamps are utilized to simulate different solar radiation levels. With forced air, the heat sink cooling system reduces the outermost temperature of PV panels by about 11 %. As a result, the PV system's efficiency increased by 16 % [104].

The cooling rate can also be influenced by the channels' position in addition to their geometry. Forced air flow was used as the cooling medium in a numerical investigation of the impact of cooling channel location on PV performance [105]. Researchers investigate two scenarios: one in which the PV cell has a cooling canal above it, and one in which it has one just below it. The study's practical and theoretical results indicate that the cooling channel above the PV was preferable due to its lower inlet air temperature [105]. In order to improve the reliability of PV modules using forced air-cooling technology, Sajjad et al. [106] has presented another study. Air is circulated using a mechanical instrument, such a pump, in this heat control strategy. The study's findings were contrasted with PV modules that did not have cooling. Comparative analysis revealed that employing the air-cooling strategy led to 7.2 % and 6 % greater electrical performance [106].

Using the wasted air from air conditioner (AC) systems, Habchi and partners exhibited a novel method for gently cooling the back surface of PV panels using nature's coolant. There is no longer a need for additional fans or increased power usage thanks to this creative technique. Furthermore, by using this novel design, warm air that would have ordinarily leaked out of the PV panel may be captured and used again for drying purposes. In order to cool the PV panels, the air conditioning system's exhaust is used. In order to do this, the air exiting the exhaust grill is directed into a nozzle, which is subsequently equipped to the PV panel. Due to heat accumulation from the surrounding air, it is anticipated that the exhaust air entering the nozzle has a temperature of 25 °C. After leaving the nozzle, the air travels through a conduit that has been specifically created to pass over the rear of the PV panel. Fiberglass insulation is used in the construction of the coolant channel's lower side in order to maximize heat transfer between the air flow and the PV panel. The findings of the computer simulations used to assess the suggested cooling system are encouraging. The PV panels' surface temperature is significantly lowered during the cooling process, increasing their electric efficiency. When there is one PV panel, the cooling system's relative enhancement ranges from 49 % to 56 %; when there are ten panels, the enhancement falls to roughly 4 %–7 %. Additionally, the PV panels' electric efficiency is increased; this gain ranges from 15 % for a single PV module to roughly 3 % for ten PV modules [107].

Salameh et al. [108] replaced a pump-dependent water-cooling system with the exhausted air produced by an HVAC (heating, ventilation, and air conditioning) system, which is commonly seen in industrial settings. Using numerical methods, they carried out a feasibility analysis and showed how a number of variables, such as the coefficient of convection and the kind of flow (forced or natural), could affect the system's output power. Three different external flow patterns were tested: the top and lower surfaces of the PV were cooled by exhausted air; a vortex generator was positioned at the leading edge of the lower surface, while the higher half was cooled by natural convection; and a vortex generator was positioned on both sides of the PV. The efficiency of the PV system increases from 11 % to 18 % when cooling demand increases up to 160 kW and solar radiation equals 500 W/m^2. To boost efficiency, turbulence flow was produced at the leading edge of the panel. When an internal flow was used on the bottom surface of the PV module, the system's efficiency increased with its length [108]. Table 5 shows studies in active air-cooling.

5. Combined PV cooling

The vast development in PV cooling method not only stop in active and passive cooling, but also the combination between both. Some papers had studied the combined PV cooling which show the different improvement in electrical efficiency and temperature reduction. PV module performance, durability, and thermal utilization may be enhanced by combining active and passive cooling techniques. Passive methods, which cool more slowly, can be paired with active methods, which cool more quickly and have greater electrical performance. It is important to combine various cooling techniques carefully since some released heat after removing it from PV arrays. The combined cooling method in other word can be the best solution and configuration for PV cooling process [21].

According to Sudakhar et al.'s findings, the PCM integrated water-cooling technique with a constant supply of water improved the power output performance of photovoltaic panels by effectively managing their thermal state, as opposed to cooling with the highest permitted temperature mechanism. The schematic configuration of the integrated PCM and water-cooling system was displayed in Fig. 16.

Performance of a photovoltaic module is significantly affected by an increase in its operating temperature. In order to solve this issue, the combination of phase change material (PCM) and an unforced water-cooling system is studied for effective PV panel heat management. OM 35 PCM was placed at the tedlar surface of the PV panel during the experiment, which involved allowing water to flow from the bottom of the PCM chamber. Different scenarios that considered the direction and duration of the cooling water flow were evaluated and compared with the reference panel. Many features, such as energy balancing, electrical and thermal efficiency, and the energy efficiency of the PV panel both with and without cooling, are compared and discussed. Studies showed that the performance of the PV panel was enhanced by the addition of PCM with natural water circulation. With an associated rise in average production of electricity, a percentage change in electrical efficiency, a

Table 5
Results in active air-cooling findings.

Reference	PV type	Cooling method specification	Temperature reduction	Electric performance
Kabeel et al. [109]	Polycrystalline PV (130 W)	Forced air cooling PV with reflectors	The temperature of PV modules is lowered to between 42.5 °C and 65.2 °C.	The results showed the improvement in electrical power from cooling PV system was 21.62 %.
[28]	Monocrystalline PV (150 W)	Air-cooling technique was used.	The result in temperature reduction was 56.36 °C–38.31 °C.	The reference and investigated solar panels have average efficiencies of 10.98 % and 11.95 %, respectively. A mean increase in electrical efficiency of 0.97 % has been seen using the suggested waste air cooling technique.
[106]	Monocrystalline PV (40 W)	Forced air cooling integrated with air-conditioner (AC)	The reason for the module's drop in temperature in the evening is a sudden drop in irradiance, which slows down the photovoltaic process and lowers module temperatures.	The efficiency of the cooling module is 7.2 % higher than that of the uncooled one.
[110]	Polycrystalline PV (250 W)	Forced air cooling supported with Buried Heat Exchanger	Panel temperature decreases to 8 °C, 10 °C, 11 °C, and 13 °C when compared to reference panel temperature.	The electrical efficiency values were 4.33 %, 9.20 %, 15.28 %, and 22.98 %

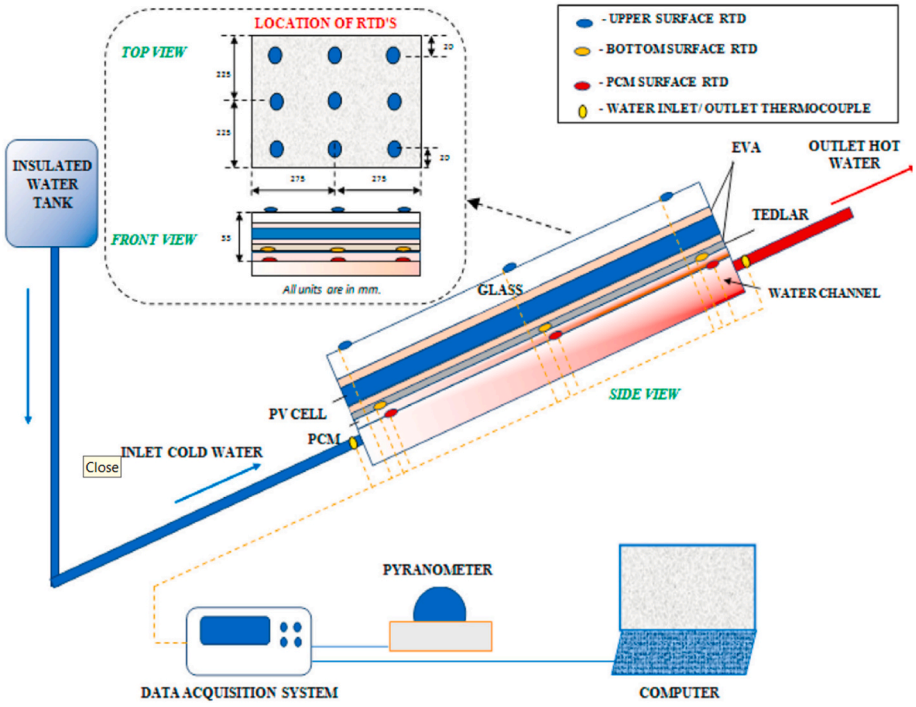


Fig. 16. PV-PCM experimental diagram integrating with water cooling [67].

reduction in the typical temperature, a maximum total energy output, and an exergy rate of 11.92 %, 12.4 %, 13.54 %, 5.4 °C, 26.07 %, and 8.08 %, respectively. It was also found that the top to bottom constant water supply cooling technique performed better than the remaining cases [67].

In a study by Agyekum and lab mates, passive and active cooling measures were coupled to boost the electrical output of a PV module. An aluminum fin heat sink and an ultrasonic humidifier were used to cool the panel. The ultrasonic humidifier was utilized to generate a humid environment behind the photovoltaic module. The study's chilling process resulted in a 14.61 °C drop in the panel's average temperature. This decrease resulted in a 6.8 % increase in the module's electrical efficiency. In contrast to the benchmark module's (10.87 W) average power, the cooled panel indicated an average power of 12.23 W. Over the course of the experiment, 1.5 L of water were approximately utilized due to evaporation. Fig. 17 shows the schematic diagram of the proposed system [111].

Yang et al. [112] proposed a cooling system that mixed geothermal energy with water spray in order to objectively evaluate the effect of decreasing temperature on the performance of PV modules.



Fig. 17. Schematic diagram of the proposed system [183].

Additionally, they created a mathematical framework to predict the system's output. The results of the mathematical framework and the experiment showed similar trends, which were verified. PV panels without cooling can have a conversion efficiency of 14.3 % lower than those with a combined water spray and geothermal energy cooling system [112]. Elminshawy et al. [110] accomplished a field test of a hybrid cooling system in a different study. The system used forced air flow and geothermal energy to enhance PV module performance and reduce temperature. The study's conclusions show that the PV module's average temperature decreased from 55 °C (without a cooling system) to 42 °C (with a combined cooling system). When this mixed cooling system was applied, the PV module's outcome power and energy output rose by an average rate of 18.90 % and 22.88 %, respectively [110].

Idoko et al. [113] developed the combined cooling technique with multi-concepts, which is a concept that incorporates passive air cooling and active water cooling. Two 250 W PV modules were used in the setup of the experiment. Both are positioned at 37 cm above the ground to allow air circulation, and water cooling was applied to one of the PV modules' surfaces to put the surface temperature down to 20 °C. The same module's rear is fastened to an aluminum fin. The module's surface cooling was aided by the water in the aluminum heat sink, which has an aluminum plate with fins connected to help with cooling. The water was cooled by adding ice blocks to the water. Utilizing an 80 % derating factor for PV array power output, the power output achieved was analyzed using the equation. The module became more productive and efficient as a result of the experiment, which reported an increase in power generation by 20.96 W and an efficiency obtained at least 3 % [113].

The starting point in research from consideration in lack of study on reusing the cooled air acquired in the evaporative cooling process, lead to a novel hybrid photovoltaic evaporative cooling system (PV/EC) that was proposed by Mahmood and Aljubury [114]. It simultaneously supplied cool air and increases the PV panel's electric power output through evaporative cooling. The impact of air velocity, water flow rate, and pad thickness on the electric power production and efficiency of PV panels was examined in this study. The groundbreaking PV/EC combined a PV panel and an evaporative cooling system into a single construction. This method decreased space compared to a PV panel-based cooling system. In addition, the new architectural structure reduces the amount of solar radiation that reaches the building's roof throughout the summer [114]. Fig. 18 shows the schematic of PV/EC.

As can be seen in the illustration above, the pad was installed directly on the back side of the PV panel, and gave better results than other studies that cooled the PV panel by evaporative cooling and left a gap between the PV panel and the pad. The suggested evaporative system tested a cellulose cooling pad with three thicknesses (50, 100, and 150 mm), three water flow rates (1, 2, and 3 LPM), and air velocity ranging from 2 to 3 m/s. PV/EC systems outperformed standard PV panels without cooling in terms of both electricity and thermal performance. PV panel efficiency increased by 7.4 %, 10.5 %, and 11.2 % for pads type 1, type 2, and type 3, respectively. Pad type 1 had an average temperature drop of 15 °C, whereas pads type 2 and type 3 were about 20 °C. The air temperature differential was 5.5 °C, 9.2 °C, and 13.9 °C for the three pads, respectively. The optimal supplied air dry-bulb temperature was 24.7 °C with 71 % relative humidity for pad type 3 [114].

Other combined cooling in water-based and air-based systems have shown to be the most practical and successful, despite the possibility of problems with thermal shock and front glass optical property loss. To get around these restrictions, Bevilacqua et al. [115] presented the results of three distinct cooling systems that were used on the rear surface of the PV module during an eight-month collection period. While the third approach makes use of the forced ventilation of a cavity beneath the module, the other two rely on a spray cooling process. The study evaluates and displays the cooling systems' operational results on the module rear surface. These systems, which may be simply installed in existing facilities, are inexpensive and simple to operate. This allowed for the evaluation of these systems' performance under various seasonal and meteorological conditions. The back-module surface spray cooling systems demonstrated their effectiveness in enhancing photovoltaic conversion, increasing monthly generated energy to 8.6 % and average total value of 6.1 %. Over the course of the analysis period, the related water usage was 2.78 m^3 [115]. Table 6 shows a comparison among discoveries in combined cooling techniques.

6. Current strategies in performance assessment of PV cooling methods

The common review on cooling PV field usually put attention in the

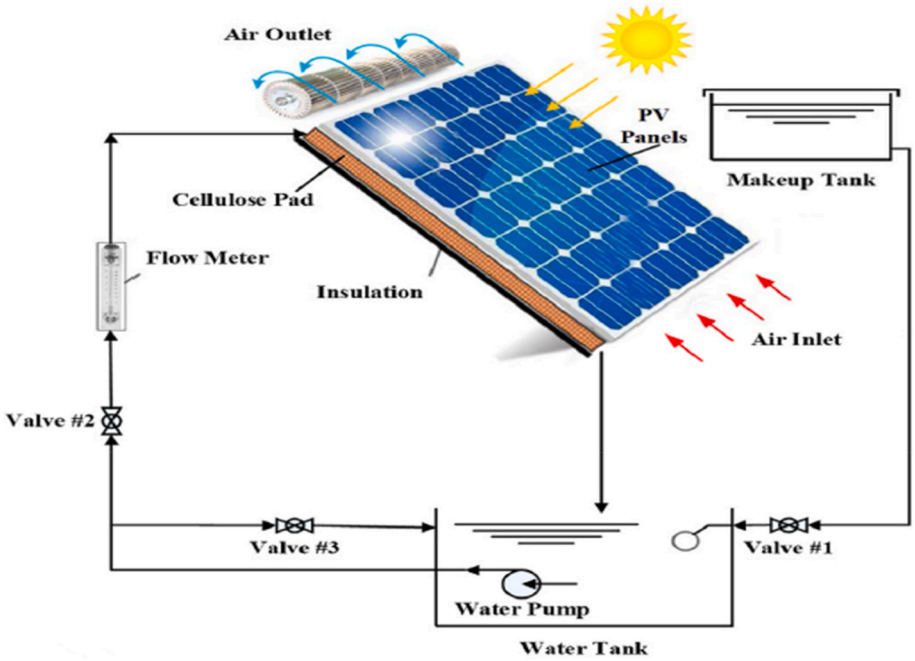


Fig. 18. Experimental diagram of PV/EC system [114].

Table 6
Comparison among discoveries in combined cooling techniques.

Reference	PV Type	Cooling method specification	Temperature reduction	Electric performance
Preet et al. [116]	Silicon based PV (100 W)	Water cooling and two absorber plates.	Temperature reduction reached about 85 °C,	Increased peak point in open circuit voltage and short circuit current by 6.75 V and 3.5 A.
[184]	Monocrystalline PV (10 W)	PCM cooling in aluminum tubes.	The PV's top surface measured 44.8 °C, whereas the bottom/back surface was 41.48 °C.	The average power and efficiency increase is 22.6 W and 21.2 %, respectively.
[185]	Crystalline Silicon Panel	Natural convection with micro fin and PCM	The temperature was lowered by 10.7 °C with PCM and 12.5 °C with PCM and micro-fins.	The results with and without fins have output power of 5.35 % and 4.8 %, respectively.
[117]	Polycrystalline PV (75 W)	Extended Fines and PCM	Throughout the day, there is a 4.7 °C shift between the upper and lower surface temperatures.	Additional 7.6 % in power output.
[109]	Polycrystalline PV (130 W)	Multi methods in combined cooling: PV reflector, forced air cooled, water cooling.	Reduced PV temperature by 19 °C with reflector and forced air cooling; and 2 °C with PV-reflector water cooling method.	The use of reflectors with cooling increased the electrical power produced by the PV module by 21.62, 35.13, and 39.69 % for multi cases respectively.
[15]	Monocrystalline PV (PV)	Water cooling with nanofluid	The percentage of PV temperature reduction is from 3.92 % to 15.26 %.	The PV panel output power rose by 12.6 %.
[118]	Crystalline PV	Water based cooling	PV temperature reduction is from 58 °C to 55 °C.	Efficiency increased by 7 %.
[114]	Monocrystalline PV (150 W)	Hybrid evaporative cooling	The average temperature drop was 15 and 20 °C	PV panel efficiency increased by 7.4 %, 10.5 %, and 11.2 %.

result and performance analysis. But the important and standard parameters to measure the performance of cooling PV methods are rarely discussed. There are many aspects that can contribute to the performance of the PV cooling such as: temperature-dependent PV efficiency difference factor (F_{TDED}), temperature dependent photovoltaic power difference factor (F_{TDPD}), PV power difference factor (F_{ED}), power ratio (R), PV cooler lifespan efficacy factor (F_{LSE}), production cost effectiveness factor (F_{CE}), modified production cost effectiveness factor (F_{MCE}), and cost and area effectiveness factor (F_{CAE}). All of these parameters have already studied and published in previous works. Table 7 presents the equations of each performance assessment method.

Each parameter identified specific condition related to the PV cooling performance and could be different of each other. In order to calculate cost in PV cooling method, there are two factors called production cost effectiveness factor (F_{CE}), and modified production cost effectiveness factor (F_{MCE}). The evaluation in terms of cost and area for PV coolers is also discussed using cost and area effectiveness factor (F_{CAE}). Certain PV cooling techniques may have greater manufacturing costs than their power productivity; therefore, it is necessary to take this cost into account in order to justify the production of such techniques.

Due to power and temperature-dependent aspect, there are four related factors. The research suggests novel techniques that are flexible enough to be used in a range of solar irradiance levels. Therefore, the cost of the assessment can be greatly reduced as a result by using temperature dependent PV power difference factor (F_{TDPD}) [188]. In the other case, a strategy for evaluating PV cooling methods was suggested, that is the temperature-dependent PV efficiency difference factor (F_{TDED}). This component determines the pertinent parameters that affect the efficiency and result in evaluating the PV cooling method as a whole [189].

Besides that, another approach is put forth to address the expense problem in performance evaluations. Under different levels of solar radiation, the PV power differential factor (F_{ED}) can be used. F_{ED} depends on the output power of a PV with a single solar cell only when evaluating the performance of a PV with a cooling device with known number of solar cells. It is demonstrated that PV cooling systems can have a significant reduction in their performance evaluation cost [121].

The aspect of power in PV module must be repeatedly calculated using the traditional technique which takes a long time since the coolers' performances must be compared using the same reference power. Consequently, a novel evaluation method was proposed, which is based on the power ratio of the temperature-dependent PV module (R). Two examples of relevant metrics found by the new technique are the power of a PV module with cooling and the reference power at the standard test conditions (STC) for PV modules, which are essential for evaluating the cooler's performance [190]. It is difficult to compare different types of PV coolers when the lifespan of the cooler is considered using the conventional method. Hence, a novel approach was proposed based on the PV cooler lifespan efficacy factor (F_{LSE}) [124]. The following sections will illustrate the above-mentioned assessment methods in details.

6.1. Production cost effectiveness factor (F_{CE})

An economic study was suggested in order to encourage the development of particular PV coolers based on their fabrication cost and power productivity. Sultan et al. suggested a metric called the PV cooling technology production cost effectiveness factor (F_{CE}). The power of a PV system with and without a cooler, the price of 1 W of PV electricity, and the cost of producing the PV cooler were used to define and determine F_{CE}. In order to determine the best kind of PV cooler, the minimum F_{CE} value must be determined. The following is a representation of the F_{CE}.

$$F_{CE} = \frac{P_{PV} + \frac{Z}{Y}}{P_{PVCT}}, \quad (1)$$

Y is the price of 1 W of PV power, while Z is the cost of the cooling technique. The output power of a PV with and without a cooler is denoted by P_{PVCT} and P_{PV}, respectively. The following is the minimal value of $F_{CE,min}$, which denotes the PV cooler's optimal performance:

$$F_{CE,min} = \frac{P_{PV}}{P_{PVCT,max}}, \quad (2)$$

Table 7
Current PV cooling strategies for performance assessment methods.

Performance Factor	Equation	Description	Reference
Production cost effectiveness factor (F_{CE})	$F_{CE} = \frac{P_{PV} + \frac{Z}{Y}}{P_{PVCT}}$	This approach correlates the manufacturing cost of the PV enhancer with the resulting power increase when the cooler is integrated into the PV.	[119]
Modified production cost effectiveness factor (F_{MCE})	$F_{MCE} = \frac{n \times P_{cell} + \frac{Z}{Y}}{P_{PVCT}}$	This approach has the potential to significantly lower the evaluation expenses associated with PV coolers.	[186]
PV cooling technique lifespan factor (F_{LSE})	$F_{LSE} = \frac{L_{PVCT}}{L_{PV}}$	The efficacy of the PV cooler's lifespan is contingent upon the longevity of both the PV cell and the cooler itself. This measure, known as the PV cooler lifespan effectiveness factor, represents the ratio of the cooler's lifespan to that of the PV cell. Employing this method aids in classifying the PV cooler's performance regarding lifespan effectiveness.	[187]
PV efficiency difference factor (F_{TDED})	$F_{TDED} = \beta_{ref}(T_{PV} - T_{PVCT}) - \frac{P_{fc}}{P_{PV,max}}$	This parameter serves as an indicator of whether the cooling method is enhancing or diminishing PV efficiency, or if it remains neutral. It could potentially serve as a metric for evaluating the performance of PV coolers by manufacturers and designers.	[120]
Modified PV efficiency difference factor (F_{TDPD})	$F_{TDPD} = \frac{I}{I_{STC}}\left[\beta_{ref}(T_{cell} - T_{PVCT})\right] - \frac{P_{fc}}{P_{PV,max}}$	This approach is adaptable and can be utilized across a range of solar irradiance levels. It relies on a PV module containing only a single solar cell, without the inclusion of a cooler.	[188]
PV power ratio (R)	$R = \frac{I}{I_{STC}}\left[1 - \beta_{ref}(T_{PVCT} - T_{ref})\right] - \frac{P_{fc}}{P_{PV,max}}$	This method instantly computes the unknown power across various reference powers, simplifying the comparison of performance among different enhancers.	[121]
PV power difference factor (F_{ED})	$F_{ED} = \frac{I_{STC}}{I}\left(\frac{P_{PVCT} - P_{fc} - n \times P_{cell}}{P_{PV,max}}\right)$	The method evaluates the performance of a PV system by comparing it to the power output of a standalone PV module with a single solar cell (without a cooler). This reference point is used to assess PV systems with varying numbers of solar cells, each equipped with a cooler. It is applicable for evaluating different types of PV coolers.	[122]
Cost and area effectiveness factor (F_{CAE})	$F_{CAE} = \left(\frac{A_{PVCT}}{A_{PV} + A_{PV,converted}}\right) \times \frac{P_{PV} + \frac{Z}{Y}}{P_{PVCT}}$	This method correlates the manufacturing cost and the area of the PV cooler with the PV power when the cooler is attached to the PV.	[123]

The maximum power output from a PV with a cooler that is comparable to the output power from a PV at STC is denoted by $P_{(PVCT,max)}$.

There are three conditions for F_{CE} based on Eq. (1).

1. The PV cooler is not likely to be cost-effective if $F_{CE} > 1$.
2. The PV cooler is deemed neutral and the threshold value is reached if $F_{CE} = 1$.
3. It is implied that the PV cooler is cost-effective if $F_{CE,min} \leq F_{CE} < 1$.

6.2. Modified production cost effectiveness factor (F_{MCE})

Sultan et al. [125] have suggested a modified economic approach to reduce the cost of PV enhancer performance evaluation. The output power of a PV with an enhancer that has a known number of solar cells and the output power of a single solar cell without an enhancer serve as the foundation for calculating the modified production cost effectiveness factor (F_{MCE}) and its minimum value. The price of producing the PV enhancer, the cost of a watt of PV power, and the output power of a PV with and without an enhancer are additional factors [125].

The modified production cost effectiveness factor (F_{MCE}) described as equation below:

$$F_{MCE} = \frac{n \times P_{cell} + \frac{Z}{Y}}{P_{PVCT}} \quad (3)$$

It is clear that F_{MCE} depends on the following items.

1. A single solar cell's (P_{cell}) power output without an enhancer.
2. The PV's output power when enhanced, or P_{PVCT}.
3. The number of solar cells that can be used in a PV with an enhancer, n.
4. The PV enhancer's manufacturing price is Z.
5. The cost per watt of PV power is Y.

The minimum value of F_{MCE} can be represented as the following formula:

$$F_{MCE,min} = \frac{P_{cell}}{P_{cell,max}}, \tag{4}$$

The solar cell's output power at STC is represented by $P_{cell,max}$. F_{MCE} conditions are comparable to F_{CE} conditions that are listed in section 6.1 earlier.

6.3. Temperature-dependent efficiency difference factor (F_{TDED})

The formulation and derivation of a unique parameter called the temperature-dependent PV efficiency differential factor (F_{TDED}) serve as a basis for a strategy for evaluating PV cooling approaches. In order to evaluate the performance of the PV cooling method, this component ascertains the relevant elements that impact efficiency. Product developers and producers can utilize this parameter as an indicator to evaluate the effectiveness of PV coolers by determining if the cooling technique is neutral, increasing, or decreasing PV efficiency [120]. The following is an illustration of F_{TDED}.

$$F_{TDED} = \beta_{ref}(T_{PV} - T_{PVCT}) - \frac{P_{fc}}{P_{PV,max}}, \tag{5}$$

where T_{PVCT} and T_{PV} represent the PV's temperature with and without a cooling method, respectively. The reference efficiency drop per unit temperature rise is represented by β_{ref}. The pumping demand for the forced circulation mode is P_{fc}. The highest output power that a PV can produce at STC is $P_{PV,max}$.

Three scenarios for F_{TDED} can be derived from Eq. (5).

1. It is implied that the cooling method adds to the increase in PV efficiency if $F_{TDED} > 0$.
2. It indicates that the cooling method is the threshold value of the gain and that it neither increases nor decreases PV efficiency if $F_{TDED} = 0$.
3. It is implied that the cooling method adds to the decrease in PV efficiency if $F_{TDED} < 0$.

6.4. Temperature dependent photovoltaic power difference factor (F_{TDPD})

There is a direct relationship between a PV's cost and quantity of solar cells; as the PV's number of solar cells rises, so does its cost. The performance becomes costly because a PV with and without a cooler require the same total number of solar cells. To put it another way, 200 solar cells are required if a PV with a cooling mechanism that uses 100 solar cells is used. This means that a PV without a cooler should have the same amounts of solar cells. Consequently, a unique method that relies on the power output of a PV with a single solar cell without a cooler is proposed in order to reduce the cost of PV cooler performance testing [124]. Furthermore, this technique is adaptable enough to be used under different solar irradiance conditions. The following is an expression for the temperature-dependent photovoltaic power differential factor, or F_{TDPD}.

$$F_{TDPD} = \frac{I}{I_{STC}}\left[\beta_{ref}(T_{cell} - T_{PVCT})\right] - \frac{P_{fc}}{P_{PV,max}}, \tag{6}$$

where I and I_{STC} stand for the solar irradiance and the 1000 W/m^2 STC solar irradiance, respectively. The F_{TDPD} conditions are comparable to the F_{TDED} conditions in section 6.3.

6.4.1. Power difference factor (F_{ED})

The efficiency difference factor (F_{ED}) is another method for lowering the cost of the PV cooler performance measurement. F_{ED} is capable of evaluating the performance of the PV coolers under varying solar irradiance levels. The output power of a PV with a single solar cell is also a factor to consider when evaluating the performance of a PV with a cooler that has an identified number of solar cells. The F_{ED} can be visualized as the following [122].

$$F_{ED} = \frac{I_{STC}}{I}\left(\frac{P_{PVCT} - P_{fc} - n \times P_{cell}}{P_{PV,max}}\right). \tag{7}$$

The equation for the natural convection cooling becomes

$$F_{ED} = \frac{I_{STC}}{I}\left(\frac{P_{PVCT} - n \times P_{cell}}{P_{PV,max}}\right) \tag{8}$$

The F_{ED} is obviously dependent on.

1. P_{cell} is the power output from one solar cell without a cooler.
2. The PV power output with a cooler is called P_{PVCT}.
3. The PV output power at STC is mentioned by $P_{PV,max}$.
4. The available solar cell count in a PV with a cooler is referred by n.
5. The pumping power is P_{fc}.
6. I is the solar irradiance.
7. I_{STC} which is the solar irradiance at 1000 W/m^2.

6.5. Power ratio (R)

The calculation consumes an extended time since the coolers' performance with natural convection must be compared using the same PV reference power. Consequently, an evaluation method is proposed, which is based on the power ratio of the temperature-dependent photovoltaic module (R). Two examples of relevant metrics that R identifies and which are essential for evaluating the cooler's performance are the power of a PV module with a cooler and the reference power at STC for PV. As a result, evaluating the performance of different coolers is made easier and the unknown power for different reference powers may be estimated instantly rather than requiring the laborious process that the usual method implies [121].

Eq. (9) indicates that R depends on the following: T_{PVCT} is the temperature of a PV system after a cooler added. T_{ref} is the reference PV temperature. β_{ref} is the reference fractional decrease in PV efficiency per unit increase in temperature. P_{fc} is the power for pumping requirement. $P_{PV,max}$ is the PV maximum output power at STC. I and I_{STC} are the incident and STC solar irradiance, respectively. Therefore, R can be represented as follows:

$$R = \frac{I}{I_{STC}}\left[1\text{-}\beta_{ref}\left(T_{PVCT} - T_{ref}\right)\right] - \frac{P_{fc}}{P_{PV,max}}. \tag{9}$$

if convection cooling occurs naturally, the equation will become as:

$$R = \frac{I}{I_{STC}}\left[1\text{-}\beta_{ref}\left(T_{PVCT} - T_{ref}\right)\right]. \tag{10}$$

6.6. Lifespan efficacy factor (F_{LSE})

The lifespan efficacy factor (F_{LSE}) is an important parameter which depends on the lifespan of the PV and the cooling technique that is used to evaluate the PV cooler's performance. In order to categorize the performance of PV coolers in terms of lifespan efficacy, the ratio of the PV cooler's lifespan (L_{PVCT}) to the PV's lifespan (L_{PV}) is known as F_{LSE} [124]. F_{LSE} can be shown in the following way.

$$F_{LSE} = \frac{L_{PVCT}}{L_{PV}}. \tag{11}$$

The following F_{LSE} criteria can be determined.

1. The PV cooler is lifespan effective if $0 < F_{LSE} \leq 1$.
2. The PV cooler reached maximum performance and longevity if $F_{LSE} = 1$.
3. The PV cooler performs the worst if $F_{LSE} = 0$.

6.7. Cost and area effectiveness factor (F_{CAE})

A metric termed the "cost and area effectiveness factor (F_{CAE})" was introduced to assess the economic feasibility of the coolers of a PV [123]. F_{CAE} was denoted as

$$F_{CAE} = \left(\frac{A_{PVCT}}{A_{PV} + A_{PV,converted}}\right) \times \left(\frac{YP_{PV} + Z}{YP_{PVCT}}\right), \tag{11a}$$

where

$$A_{PV,converted} = \frac{P_{PV,enhanced} \times A_{PV}}{P_{PV}}, \tag{12}$$

where

$$P_{PV,enhanced} = P_{PVCT} - P_{PV}. \tag{13}$$

$P_{PV,enhanced}$ is the additional power generated by a PV system when cooled by a specific device. Z represents the manufacturing cost of the cooler. $P_{PV,out}$ denotes the power output of a standard PV system without a cooler. P_{PVCT} signifies the power output of a PV system with the cooler. A_{PV} and A_{PVE} refer to the surface areas of the standard PV system and the enhanced PV system, respectively. $A_{PV,converted}$ is the converted area from the enhanced power using an enhancer. Y represents the cost per watt of PV power. F_{CAE} conditions are comparable to F_{CE} conditions that are listed in section 6.1.

7. Comparison between existing and current literature reviews

In this review, the classification of the cooling PV will be stated in three broad types of passive cooling, active cooling, and combined cooling. Each type is explained with examples. Additionally, a performance comparison of these cooling techniques is provided to illustrate the advancements made in this field of study in recent years. It also includes the evaluation methodology for every kind of cooling technique, providing a lucid mathematical and physical framework for this field of study. This work offers a thorough review of earlier studies that explore different strategies for enhancing the cooling PV technique performance within this framework. The three primary subfields of cooling technique are explored, compared, and analyzed. Table 8 shows a compassion between the existing and the current literature review.

8. Conclusion

Typically, previous discussions on PV cooling focus on analyzing the performance. However, there's a lack of emphasis on discussing the crucial and standard parameters essential for assessing the effectiveness of cooling methods for PV systems. In this review article, the classification of cooling PV with several updated researches and works in PV cooling technology are discussed. The classification of cooling PV are divided into three main groups: passive cooling, active cooling, and combined cooling. Passive cooling consists of fin cooling, Phase Change Materials (PCMs), Thermosyphon, Thermoelectric Generator (TEG). Besides that, active cooling: Active water cooling and air cooling. Meanwhile, combined cooling lime PCM with water cooling or air-water cooling. On the other hand, the assessment methods to evaluate the performance of the PV cooling in different aspects are also discussed, such as economic, power, efficiency, lifespan, and etc. These assessment methods are PV cooling technique lifespan factor (F_{LSE}), PV efficiency difference factor (F_{TDED}), modified PV efficiency difference factor (F_{TDPD}), PV power ratio (R), and power difference factor (F_{ED}),

Table 8
Comparison among review articles in PV cooling technology.

Reference	Review Framework	PV cooling classification	Performance assessment method description
[21]	The study assesses the sophisticated cooling techniques used in photovoltaic (PV) and concentrated photovoltaic (CPV) modules in terms of performance and environmental factors. The study reviewed and categorized the cooling techniques based on a number of research articles, study focus, research host countries, technology type, cooling effect, and electrical efficiency enhancement.	Active, passive, and combined cooling.	No description.
[29])	This research provides an in-depth analysis of PV cooling. Enhancing the life span and electrical efficiency of PV is the goal of the critical examination. Several benefits and drawbacks of various methods in active and passive cooling are also covered.	Active and passive cooling.	No description.
[27]	This study provides a thorough analysis of previous research that covers a wide range of strategies for improving performance by optimizing PV operating temperature and minimizing the detrimental effects of rising temperatures on PV. Additional cooling methods for PV employing PVTS (Photovoltaic Thermal-based Systems) was provided. To guarantee reduced PV deterioration, research has been done on air-based, water-based, and hybrid cooling systems that use phase change material (PCM), nanomaterials, and nanofluids. Reports on the efficiency of each cooling strategy and PV's performance improvement was given.	Water cooling, air cooling, fin, hybrid methods (PCM, fin, nanoparticle, and nanofluids)	No description.
[8]	The report compiles and compares the substantial research done by scientists using passive cooling methods. Various PV passive cooling methodologies such as natural ventilation,	Passive cooling: natural ventilation, liquid immersion or floating, heat pipe, PCM, spectral beam splitter, and thermoelectric.	No description.

(continued on next page)

Table 8 (*continued*)

Reference	Review Framework	PV cooling classification	Performance assessment method description
	liquid immersion or floating, heat pipe, thermoelectric, PCM, and spectral beam splitter are discussed. For each passive method, some papers are presented and supplementary recommendations are provided.		
[24]	The goal of the paper is to present a thorough and current analysis of sophisticated cooling methods for solar systems, emphasizing the critical role these methods play in raising the sustainability and efficiency of these systems. The study starts with a succinct theoretical overview of PV technology and the key influencing elements that affect it, highlighting the vital role that cooling technologies play in enhancing system performance as a whole. Explanatory graphics that clarify the fundamentals of each cooling strategy are included in the review, which neatly arranges a variety of cooling approaches in an understandable and straightforward manner. In addition, a thorough examination, debate, and contrast of these methods according to the performance gains that each yield are carried out in order to offer important suggestions for further study and advancement in this area. This study is unique because it takes a methodical approach to categorizing and assessing various cooling methods, which has made a substantial contribution to the advancement of PV cooling methods.	Phase-change materials, air and water cooling, and a variety of other active and passive cooling techniques.	No description
[7]	A thorough study of Water Photovoltaic (WPV) systems has been carried out in this paper. Shallow water (lakes, ponds, dams) PV type deployments are becoming more and more common among	WPV: Floating, underwater, offshore and canal top	No description

Table 8 (*continued*)

Reference	Review Framework	PV cooling classification	Performance assessment method description
	WPV. Planning in advance is therefore crucial in order to reduce natural disaster damage and enhance the environmental effect prior to installing a WPV system.		
[22]	The article includes an overview and evaluation of active and passive temperature management techniques for photovoltaic technology. This research focuses on the design and performance factors of PV using fins. Fins are utilized in photovoltaic thermal system (PVT) for air, fluid, nanofluid, and PCM cooling. By installing them in these systems, their performance is enhanced and their combined electrical and thermal efficiency is increased at a rate of 8–10 %. This article also looks at several methods for developing and putting into practice systems, optimizing outcomes, and investigating opportunities and potential issues. The goal of this article is to develop, produce, use, and promote high-performance PV technologies and systems that incorporate fins.	Fin heat sink.	No description
Current study	In this review article, several updated studies in PV cooling technology are discussed including passive, active and combined cooling. Besides that, the performance assessment methods are also elaborated. It was found that the PV cooling has some major challenges to be addressed in future studies including cost, maintenance requirement, energy consumption especially for active cooling methods.	Passive, active and combined cooling	The common reviews on PV cooling usually put attention in the result and performance analysis. But the important and standard parameters to measure the performance of cooling PV methods is rarely discussed. There are many aspects that can contribute to the performance of the PV cooling, such as: temperature-dependent PV efficiency difference factor (F_{TDED}),

(*continued on next page*)

Table 8 (*continued*)

Reference	Review Framework	PV cooling classification	Performance assessment method description
			temperature dependent photovoltaic power difference factor (F_{TDPD}), PV power difference factor (F_{ED}), power ratio (R), PV cooler lifespan efficacy factor (F_{LSE}), production cost effectiveness factor (F_{CE}), modified production cost effectiveness factor (F_{MCE}), and cost and area effectiveness factor (F_{CAE})

production cost effectiveness factor (F_{CE}), modified production cost effectiveness factor (F_{MCE}), and cost and area effectiveness factor (F_{CAE}). Hence, this review paper suggests some potential future works for PV cooling as the following.

1. Advanced passive cooling techniques: Passive cooling methods, such as incorporating novel materials with high thermal conductivity or designing innovative heat sinks, could be further developed to improve heat dissipation without the need for additional energy input.
2. Active cooling systems: Enhanced active cooling systems, such as liquid cooling or thermoelectric cooling, may become more prevalent. These systems actively remove heat from PV modules, thereby improving their efficiency and longevity.
3. Smart control systems: Implementing smart control systems that can dynamically adjust cooling strategies based on environmental conditions, module temperature, and energy production requirements can optimize cooling efficiency and energy usage.
4. Integration with building design: Integrating PV cooling technologies into building design, such as incorporating cooling elements into building materials or designing structures that promote natural airflow around PV panels, could enhance overall energy efficiency and reduce cooling loads.
5. Hybrid solutions: Combining multiple cooling techniques, such as passive and active cooling or integrating PV modules with other renewable energy systems like wind turbines or solar thermal collectors, could offer synergistic benefits and improve overall system performance.
6. Nanotechnology: Continued research into nanomaterials and nanotechnology could lead to the development of coatings or surface treatments that enhance heat dissipation and improve the thermal performance of PV modules.
7. Machine learning and AI: Utilizing machine learning algorithms and artificial intelligence for predictive modelling and control of PV cooling systems can optimize energy usage, anticipate maintenance needs, and adapt to changing environmental conditions in real-time.
8. Efficiency and cost optimization: Future developments should focus on improving the efficiency and cost-effectiveness of PV cooling technologies to make them more accessible and widely adopted in both utility-scale and distributed PV systems.

By focusing on these areas of innovation and development, the future of photovoltaic cooling holds the potential for significant improvements in energy generation efficiency, reliability, and sustainability.

CRediT authorship contribution statement

Ihsan Okta Harmailil: Resources, Data curation, Formal analysis, Writing – original draft, Methodology, Investigation, Validation, Visualization. **Sakhr M. Sultan:** Conceptualization, Resources, Data curation, Formal analysis, Writing – original draft, Methodology, Investigation, Validation, Visualization, Supervision, Project administration, Funding. **C.P. Tso:** Funding acquisition. **Ahmad Fudholi:** Supervision, Resources. **Masita Mohammad:** Supervision, Resources. **Adnan Ibrahim:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

The authors would like to express their appreciation to Universiti Kebangsaan Malaysia for providing research funding No. GGPM-2023-028.

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